# ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician (Mar Tech)

USQ – Senior Engineering Watchkeeper (SEWK) HFX (ALVA)

MITE Course IDs
Residential Phase – 122018
OJPR – 122045

MARPAC RDIMS 1249309
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RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Training Group (NTG) and the Commanding Officer of the Curriculum Control Authority (CCA), Naval Fleet School (Pacific) (NFS(P)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

| Revision | Approval Date | Authority |
|---|---|---|
| 1.0 (QSPWB Month Year) | | NTDC(P) CO |
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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training Group (CNTG) and in accordance with the following references:

A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);
Annex C – 5555-00379-MAR-TECH dated 15 Dec 16;
NAVORD 4500-0 - Structured Delegation; and
NTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NTDC(P) MS Div), who will forward to NPTG for approval, as appropriate.

5. The designated TEs for this training are indicated below:

| Type | Phase / Module | MITE Course ID | NQual |
|---|---|---|---|
| Residential | | 122018 | ALVA |
| OJPR | | 122045 | ALVA |

# **CHAPTER 1** TRAINING MANAGEMENT DETAILS

## AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL TRAINING GROUP (CNTG)

**AIM**

6. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the MARTECH Senior Engineering Watchkeeper. This QSP is designed to assist the training system in developing, conducting, evaluating and validating the training.

COURSE DURATION

7. Course duration is calculated in accordance with NTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP.

| **Allocation of ATT** | **Minutes** |
|---|---|
| Lesson Objectives (LOs) / Teaching Points | 4290 |
| Enabling Checks (ECs) | 1440 |
| Performance Checks (PCs) | 720 |
| **ATT Sub-total** | **6450** |
| **Allocation of BAT, TAT, TTT, PT** (Course > 31 days) | |
| Base Administrative Time (BAT) | 120 |
| Training Administrative Time (TAT) | 120 |
| Training Travel Time (TTT) | 0 |
| Physical Fitness (PT) | 540 |
| **Sub-total** | **780** |
| **Course Duration Calculation (ATT+BAT+TTT+TAT+PT)**<br>**Total mins/400 = # days** (rounded up) | |
| **Total Training Days** | **18 days** |

RESOURCES

8. The following table details key personnel required to conduct the training:

| Required Course Staffing | | | | |
|---|---|---|---|---|
| Position | Rank | Qualifications Required | Qty | PO / EO |
| Sr. Instructor | PO1 | Sr. Engineering Watchkeeper Qualified (Legacy Cert 3E) | 1 | N/A |
| Instructor | PO2 | Sr. Engineering Watchkeeper Qualified (Legacy Cert 3E) | 1 | PO 001 |
| Instructor | MS | Sr. Engineering Watchkeeper Qualified (Legacy Cert 3E) | 1 | PO 001 |

9. The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP.

**COURSE CAPACITY**

| Critical Training Resources | PO/EO |
|---|---|
| DWAN Laptop (each trainee) | PO 001 |
| DWAN drops (each trainee) | PO 001 |
| MCRT | PO 001 |
| PCOT | PO 001 |
| Halifax class ship | PO 001 |
| OBTS | PO 001 |
| References | PO 001 |

10. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Justification can be found in Annex C. The instruction and assessments are designed to accommodate the maximum load of 12 candidates.

a. Maximum – 12;

b. Minimum – 4; and

c. Optimum – 8.

MITE

11. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

12. Scheduling. Each session of this course shall be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

13. Prerequisites. The loading authority must ensure that trainees have the following prerequisites before being selected for this course:

a. USQ S1 Mechanical Specialist (MITE 122019/122969 – NQual ALUS) or QL-5 Legacy (Mar Eng, HTECH, ETECH);

b. USQ-Engineering Watchkeeper HFX (MITE 121182/121183 – NQual ALPI) or Cert 2E (MITE 108747 – NQual AFOY);

c. Must have experience as an Engineering Watchkeeper or Cert 2E (500 hrs of console time + 180 Sea Days and a 12-month period of experience);

d. MS – PO1; and

e. Primary Leadership Qualification.

14. Qualification. Successful completion of this course and associated OJPR constitutes the primary means of attaining the USQ– Senior Engineering Watchkeeper HFX (National Qualification(s) ALVA). Completion of the OJPR is mandatory to achieve the SEWK qualification (NQUAL = ALVA).

FUELLING I/C CHECKLIST

15. To be employed as a Fuelling I/C aboard a HALIFAX Class frigate, SEWK graduates are required to be compliant with SSOs, SEMs and Fueling Orders. To ensure compliance with those Orders, SEWK graduates shall complete the Fueling I/C Checklist at Annex E – Appendix 1 of this QSP.

16. SEWK graduates are not required to complete the Fuelling I/C Checklist to receive the SEWK qualification (ALVA) or challenge the SEWK Board.

COURSE REPORTING

17. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

18. PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces Military Equivalencies Program”, and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment”.

# **CHAPTER 2** ASSESSMENT OF TRAINEE

## AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

## GENERAL

19. This is a pass/fail course. Trainees must successfully complete the PO to pass this course. Performance Checks (PCs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

20. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K for details.

21. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

## ENABLING AND PERFORMANCE CHECKS

22. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

23. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES Vol. 6 with the unsatisfactory progress documented in their Trainee File.

24. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

25. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

26. All PC failures and supplemental failures are to be handled IAW NTORD 4500-54 “Training Review Board Process”. Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

## RESPONSIBILITY FOR TESTING

27. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Learners”.

28. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NFS(P) Standards Div.

## PROGRESS MONITORING

29. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (Such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.

30. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

31. Continuous monitoring of candidate progress is required to provide the following:

a. Early warning of trainee difficulties;
b. Feedback on the effectiveness of training;
c. Information for any required TRBs; and
d. Information to assist in completion of the CF 377 Course Report.

32. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

33. Instructional staff shall maintain individual trainee files, including the results of all:

a. Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);
b. Formative assessment material (e.g. ECs, practice periHalocarbon, homework, assignments and quizzes, etc.); and
c. Summative assessment material (PCs, CECs).

## TRAINING REVIEW BOARD

34. TRBs shall be convened and conducted IAW with NTORD 4500-54 “Training Review Board Process”.

35. In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis, the necessity to convene a TRB.

## COURSE REPORTS

36. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NTORD 4500-51 “Course Reports and Certifications”.

37. Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

38. The Course Report narrative comments could include:

a. Ability to work with others;
b. Ability to work under pressure;
c. Written and verbal communication skills; and
d. Organizational skills. Leadership ability. Training limitations and / or incomplete training must be included in the course report.

# **CHAPTER 3** CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

## AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

**PO 001 - ADVISE COMMAND ON MARINE SYSTEMS OPERATION CAPABILITIES AND LIMITATIONS**

1. Performance. Advise command on marine systems operational capabilities and limitations (CT0020).

2. Conditions.

Given:

(1) References;

(2) Assistance;

(3) Supervision; and

(4) Halifax class ship.

Denied: Nil.

Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Senior Engineering Watchkeeper shall independently advise command on marine systems operational capabilities and limitations outside of normal design intent and in emergency and/or battle damage events, IAW refs by:

e. Advising command on propulsion capabilities and limitations (CT0040);

f. Advising command on machinery/system status (GT0010);

g. Advising command on Power Generation and Distribution systems (CT0030); and

h. Advising command on battle damage survey results (IT0010).

4. References. Specified in applicable EOs.

5. Task Numbers. CT0020, CT0040, CT0010, CT0030, CT0050, CT0060, CT0310,CT0320, CT0330, CT0340, GT0010, HT0070, IT0010, and IT0120.

6. Training Limitations.

i. Realism:

(1) Will not be able to realistically simulate the stress that occurs during an event;

(2) Cannot simulate the dynamic human interactions and cohesive teamwork and supervisory experience that is developed over time;

**PO 001 - Advise command on marine systems operational capabilities and limitations (CT0020).**

**EO 001.01 - ADVISE COMMAND ON PROPULSION CAPABILITIES AND LIMITATIONS**

39. Performance. Advise command on propulsion capabilities and limitations.

40. Conditions.

Given:

(1) Assistance;

(2) References;

(3) Operational IPMS;

(4) EHM; and

(5) Supervision. Interesting,

Denied: Nil.

Environmental:

(1) Classroom (DWAN); and

(2) Trainer (MCRT).

41. Standard. The trainee shall independently advise command on propulsion capabilities and limitations IAW refs by:

Performing Sr. EWK duties and responsibilities IAW A46 and D1 by:

(1) Interpreting bearing flow chart;

(2) Completing divers down chit at sea;

(3) Managing the quiet running state, to include:

(a) Purpose; and

(b) Engineering considerations.

(4) Managing notice for power (NFP) by:

(a) Addressing issues/maintain NFP at sea/alongside.

Operating Equipment Health Monitoring (EHM) to include:

(1) Navigating EHM software;

(2) Operating remotely EHM;

(3) Analyzing EHM signals by; and

(4) Utilizing event viewer.

Operating in normal/abnormal drive modes IAW ref A to include:

(1) Describing operating restrictions and parameters to include:

(a) Conducting single shaft operations to include;

(i) Locked shaft; and

(ii) Zero/flat pitch and trailing.

(b) Conducting fixed pitch operations;

(c) Steering by main engines; and

(d) Transitioning between drive modes.

Identifying parameters and operational requirements to include:

(1) Towing another vessel; and

(2) Being towed by another vessel.

Operating critical propulsion equipment/components in manual control to include:

(1) Conducting pawl free operations;

(2) Operating the clutch control cabinet;

(3) Operating the thermostatic control valve;

(4) Completing air turbine testing; and

(5) Operating all GT start permissives IAW Ref A2.

42. Time Required:

EO – 1740 mins to deliver training;

EC – 360 mins (total for all trainees); and

PC – 720 mins (total for all trainees).

43. Remarks. Nil.

44. Teaching Points.

| Teaching Points | Ref | Method / Media / Environment / Training Activity Details | Time (mins) |
|---|---|---|---|
| **LEARNING OBJECTIVE: Defining Sr. EWK duties and responsibilities by:** | | | |
| 1. Perform Sr. EWK duties and responsibilities. | A30<br>A46<br>D1 | **Method / Media:** IL.<br>**Trg Environment:** Classroom.<br>**Activity Details:**<br>Instructor to detail the duties and responsibilities of Sr EWK IAW ref via lecture. Material to be confirmed in a short (5 min) quiz on the first available class next day.<br>**Trg Resources:**<br>Duties and responsibilities handout. | 30 min |

| | | | |
|---|---|---|---|
| | A46<br>D1<br>D3 | **Method / Media:** IL, Independent study<br>**Trg Environment:** Classroom<br>**Activity Details:**<br>Trainees will be provided a DWAN laptop and an assignment sheet. They will be required to conduct a publication/reference search, locating the required material and indicating its source/location. Trainees provided 40 minutes to complete, with 10 minutes provided for instructor to correct answers in class.<br>**Trg Resources:**<br>Assignment sheet, answer sheet | 50 min |
| 2. Interpreting bearing flow chart. | A30<br>D1 | **Method / Media:** IL/DP.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor to provide lecture to trainees on how to interpret the bearing flow chart. Trainees will be provided a bearing flow chart complete with scenario and will be required to brief command (Instructor) with their recommendations. Feedback to consist of clear, accurate, and concise flow of information.<br>**Trg Resources:** Scenario and bearing flow chart. | 100 min |
| 3. Interpreting bearing flow chart. | A30<br>D1 | **Method / Media:** IL/DP.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor to provide lecture to trainees on how to interpret the bearing flow chart. Trainees will be provided a bearing flow chart complete with scenario and will be required to brief command (Instructor) with their recommendations. Feedback to consist of clear, accurate, and concise flow of information.<br>**Trg Resources:** Scenario and bearing flow chart. | 100 min |
| 4. Completing divers down chit at sea. | A56 | **Method / Media:** IL.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor to deliver lecture on procedure to follow during diving operations. Focus to be placed on the safety of diving personnel and responsibility of Sr EWK.<br>**Trg Resources:** Sample checklist. | 30 min |
| 5. Managing the quiet running state, to include:<br>a. Purpose; and Engineering considerations. | D3 | **Method / Media:** IL.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor to deliver lecture on procedure to follow when assuming different quiet running states. Focus to be placed on necessary deviations from the quiet running bill and responsibility of Sr. EWK.<br>**Trg Resources:** Sample checklist. | 30 min |
| 2. Managing notice for power (NFP) by:<br>a. Addressing issues/maintain | A46<br>D1 | **Method / Media:** GD.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor to lead a guided discussion on issues that may arise in | 50 min |

| | | | |
|---|---|---|---|
| NFP at sea/alongside. | | achieving/maintaining NFP. Focus to be placed on assumption of risk and importance of early communication.<br>**Trg Resources:** GD set up. | |
| **LEARNING OBJECTIVE: Operating Equipment Health Monitoring (EHM)** | | | |
| 1. Operating EHM remotely to include:<br>a. Log in; and<br>b. Accessing database | A21<br>A23 | **Method / Media:** Time/Stop DP (EDI).<br>**Trg Environment:** PCOT.<br>**Activity Details:** Instructor will demonstrate each step with an explanation. Trainees will practice each step and receive feedback from the instructor.<br>**Trg Resources:** PCOT with multiple ship databases. Aide memoire. | 50 min |
| 2. Navigating EHM software to include:<br>a. Identifying relevant signals;<br>b. Selecting relevant signals;<br>c. Formatting data; and<br>d. Displaying data. | A21 A23 | **Method / Media:** Time/Stop DP (EDI).<br>**Trg Environment:** PCOT.<br>**Activity Details:** Instructor will demonstrate how to identify and select relevant signals, format and display data. Trainees will perform each task and receive instructor feedback. Instructor will provide a scenario for trainees to identify relevant signals and provide an analysis and brief command (instructor) from printout to the class.<br>**Trg Resources:** PCOT with multiple ship databases. Aide memoire, scenario. | 700 min |
| 3. Analyzing EHM signals by;<br>a. Comparing current data to identify issues and/or failures. | | | |
| 4. Utilizing event viewer to include:<br>a. Compare data to identify issues and/or failures of the event. | A21<br>A23 | **Method / Media:** Time/Stop DP (EDI).<br>**Trg Environment:** PCOT.<br>**Activity Details:** Instructor will demonstrate how to identify relevant events. Trainees will perform each task and receive instructor feedback. Instructor will provide a scenario for trainees to identify relevant events and provide an analysis from printout and brief command (instructor) and the class. | 50 min |
| **LEARNING OBJECTIVE: Operating in normal/abnormal drive modes** | | | |
| 1. Describing operating restrictions and parameters to include:<br>a. Conducting single shaft operations to include;<br>i. Locked shaft; and | A2 | **Method / Media:** IL, video, mock-ups.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor will give a lecture, play videos to describe single shaft operations to including locked shaft, zero/flat pitch and trailing, fixed pitch operations including steering by main engines and transitioning between drive modes. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour until TEL is developed. | 50 min |

| | | | |
|---|---|---|---|
| ii. Zero/flat pitch and trailing.<br>b. Conducting fixed pitch operations;<br>c. Steering by main engines; and<br>d. Transitioning between drive modes. | | | |
| 2. Identifying parameters and operational requirements for:<br>a. Towing; and<br>b. Being towed. | A2 | **Method / Media:** IL, video, mock-ups.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor will give a lecture, play videos to describe all procedures associated with towing operations to include parameters and operational requirements for towing; and being towed Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour until TEL is developed. | 50 min |
| 3. Operating critical propulsion equipment/components in manual control to include but not limited to:<br>a. Pawl free;<br>b. Clutch control cabinet;<br>c. Thermostatic control valve;<br>d. Air turbine testing; and<br>e. All GT start permissives. | A2 | **Method / Media:** IL, video, mock-ups.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor will give a lecture, play videos to describe all procedures associated with manual control of Pawl free operation, Clutch control cabinet, Thermostatic control valve, Air turbine testing; and All GT start permissive. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br><br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment,(manual control procedures) mock ups, ships tour until TEL is developed | 50 min<br><br>400 min (tour) |

45. Test Details.

a. EO 001.01 will be assessed by EC 001.01 and PC 001 IAW EC 001.01 Checklist and Annex K.

**EC 001.01 - ADVISE COMMAND ON PROPULSION CAPABILITIES AND LIMITATIONS**

**EC Time:** 15 mins (per trainee) + 15 min debrief (360 minutes).

**ASSESSMENT ACTIVITY**: Trainees will be given a scenario involving an abnormal propulsion condition. Trainees will be required to identify the condition, formulate options using the EHM and provide recommendations. Trainee will advise command on options. The scenario will begin after IAs are completed by EWK.

**Pass/fail criteria**: 75% on all non-critical items. All critical items must be completed correctly.

**ASSESSMENT RESOURCES:**

- EHM;
- Scenario (various scenarios for each trainee);
- References;
- MCRT; and
- Assistance of EWK.

**IMPACT OF FAILURE:** Remedial Training / Re-test.

| Senior Engineering Watchkeeper – EC 001.01 Assessment Checklist | | | |
|---|---|---|---|
| **Candidate Name:** | **Assessor Name:** | **Date:** | **Session:** |

**INVIGILATOR NOTES:**

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 75% SATs on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.
3. Instructor act as senior engineering personnel/command.

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| **1. Duties and Responsibilities of Sr. EWK** | | | |
| Check engineering log for the following | | | |
| Ongoing maintenance; | | | |
| Defects; | | | |
| Running equipment; and | | | |
| Update the engineering log as necessary. | | | |
| Perform page rounds to include: | | | |
| Running engine data page; | | | |
| Non-running engines start permissive; | | | |
| Gearing pages; | | | |
| Shafting pages; and | | | |
| Event viewer. | | | |
| Communicate to the MCR personnel and OOW* | | | |
| Assume control of the watch. | | | |
| **2. Operating in normal/abnormal drive modes** | | | Begin scenario with a propulsion problem from EO 001.01. |
| Ensure EWK conducts IA to include: | | | |
| Responding to main propulsion emergencies; and* | | | |
| Responding to main propulsion alarms and faults;* | | | |
| Conduct an initial situation brief to the officer of the watch to include:* | | | |
| Outlining the situation;* | | | |
| Impact; and* | | | |
| Options.* | | | |
| **3. Operating Equipment Health Monitoring (EHM)** | | | |
| Navigating EHM software to include: | | | |
| Identifying relevant propulsion signals; | | | |
| Selecting relevant propulsion signals; | | | |
| Formatting propulsion data; and | | | |
| Displaying propulsion data. | | | |

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| Analyzing EHM propulsion signals by: | | | |
| Comparing current data to identify issues and/or failures; and | | | |
| Compare data in event viewer to identify issues and/or failures of the event. | | | |
| **4. Advise command on propulsion capabilities and limitations** | | | |
| Report deficiencies to senior engineering personnel.* | | | |
| Develop a plan considering limitations and restrictions based on scenario. | | | |
| Update officer of the watch with recommendations to include: | | | |
| Update on the situation; | | | |
| Options; | | | |
| Impact; | | | |
| Estimated time of repair; and | | | |
| Brief is clear and concise. | | | |
| Brief the watch on the following: * | | | |
| Safety precautions;* | | | |
| Required repairs; and | | | |
| Operational hazards.* | | | |

| ADDITIONAL COMMENTS | |
|---|---|
| **STRENGTHS:** | **AREAS FOR DEVELOPMENT:** |

| THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT | | | |
|---|---|---|---|
| **OVERALL ASSESSMENT (circle one)** | | **FAIL** | **PASS** |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

**PO 001 - Advise command on marine systems operational capabilities and limitations**

**EO 001.02 - ADVISE COMMAND ON MACHINERY/SYSTEM STATUS**

46. Performance. Advise command on machinery/system status.

47. Conditions.

Given:

(1) Assistance;

(2) References;

(3) Operational IPMS;

(4) EHM; and

(5) Supervision.

Denied. Nil.

Environmental:

(1) Classroom (DWAN); and

(2) Trainer (MCRT).

48. Standard. The trainee shall independently advise command on machinery/system status and environmental emergencies IAW refs by:

Reporting on marine systems engineering parameters and machinery limitations to include:

(1)Main Lube Oil (MLO);

(2)Controllable Reversible Propeller Pitch (CRPP);

(3)Gas turbine gas path;

(4)Propulsion Diesel Engine (PDE) airflow;

(5)Fuel Oil Services (FOS);

(6)Steering;

(7)High Pressure Air (HPA); and

(8)Low Pressure Air (LPA).

Reporting on environmental impacts to include:

(1) Halocarbon release;

(2) Fuel spill within and outside of ship; and

(3) HAZMAT spill within and outside of ship.

49. Time Required:

EO –500 mins to deliver training;

EC – 360 mins (total for all trainees); and

PC – 720 mins (total for all trainees).

50. Remarks. Nil.

51. Teaching Points.

| Teaching Points | Ref | Method / Media / Environment / Training Activity Details | Time (mins) |
|---|---|---|---|
| **LESSON OBJECTIVE: Describing Operating Restrictions and Parameters** | | | |
| 1. Describing operating restrictions and parameters to include:<br>a. Main Lube Oil (MLO):<br>i. Manual operation of thermostatic control valve;<br>ii. Manual operation of pressure regulator;<br>iii. Changing MLO filters;<br>iv. Manual operation of three way valves;<br>v. Cold Moves; and<br>vi. Turning Shafts in Dry Dock.<br>b. Controllable Reversible Propeller Pitch (CRPP) to include:<br>i. Cross connecting system; and<br>ii. Stand by pump testing and operation. | A2<br>A31<br>C1 | **Method / Media**: IL, video, mock ups.<br>**Trg Environment**: Classroom.<br>**Activity Details**: Instructor will give a lecture, play videos to describe all procedures associated with MLO. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour, operational OBTS until TEL is developed. | 50min |
| | A31<br>A35<br>A37 | **Method / Media:** IL, video, mock-ups.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture, play videos to describe all procedures associated with CRPP. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour, operational OBTS until TEL is developed. | 50min |
| c. Gas turbine gas path:<br>i. Identify surge and stall conditions;<br>ii. Operation of air intake auto bypass;<br>iii. Apply full power parameters; and<br>iv. Operation of lube oil storage and conditioning assembly (LOSCA). | A2<br>A5<br>A16<br>A31<br>C41 | **Method / Media**: IL, video, mock-ups.<br>**Trg Environment**: Classroom.<br>**Activity Details**: Instructor will give a lecture, play videos to describe all procedures associated with gas turbine gas path. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources**: Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour, operational OBTS until TEL is developed. | 50min |

| | | | |
|---|---|---|---|
| d. Propulsion Diesel Engine (PDE) airflow to include:<br>i. Explain charge air flow; and<br>ii. Manually operate by pass. | A2<br>A23<br>A31<br>A41<br>A49 | **Method / Media**: IL, video, mock-ups.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture, play videos to describe all procedures associated with PDE. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment,(manual control procedures) mock ups, ships tour, operational OBTS until TEL is developed. | 50min |
| e. Fuel Oil Services (FOS) to include:<br>i. Manual operation of fuel pressure regulators;<br>ii. Supply fuel to main machinery by centrifuge; and<br>iii. Apply safety precautions for fuel separator. | A13<br>A14<br>A31<br>A50<br>A31<br>A38<br>A39 | **Method / Media:** IL, video, mock ups.<br>**Trg Environment:** Classroom.<br>**Activity Details:** Instructor will give a lecture, play videos to describe all procedures associated with FOS. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour Operational OBTS until TEL is developed. | 50min |
| f. Steering to include:<br>i. Brief safety requirements for high pressure hydraulics. | A3<br>A4<br>A31<br>A51 | **Method / Media**: IL, video, mock ups.<br>**Trg Environment**: Classroom.<br>**Activity Details**: Instructor will give a lecture, play videos to describe all procedures associated with steering. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour, until TEL is developed. | 50min |
| g. High Pressure Air (HPA) to include:<br>i. Brief high pressure air safety requirements;<br>ii. Brief the action state of HPA;<br>h. Low Pressure Air (LPA) to include:<br>i. Brief low pressure air safety requirements; and | A31<br>A51 | **Method / Media**: IL, video, mock ups.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture, play videos to describe all procedures associated with HPA/LPA. Trainees will be given the opportunity to practice on equipment during ships tour or with mock ups.<br>**Trg Resources:** Lecture, Pictures/videos of relevant equipment, (manual control procedures) mock ups, ships tour, operational OBTS until TEL is developed. | 100 min |

| | | | |
|---|---|---|---|
| ii. Brief action state of LPA. | | | |
| **Learning Objective: Reporting on environmental impacts** | | | |
| 2. Describe actions during environmental emergencies to include:<br>a. HALOCARBON release;<br><br>b. Fuel spill within and outside of ship; and<br><br>c. HAZMAT spill within and outside of ship. | D2 | **Method / Media**: IL.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture, to describe all procedures associated with actions and reporting environmental emergencies. Trainees will be given a scenario will be required to advise command during environmental/health impacts and clean up procedures.<br>**Trg Resources:** Lecture, pictures of relevant equipment. | 100 min |

52. Test Details.

a. EO 001.02 will be assessed by in EC 001.02 and PC 001 IAW EC 001.02 Checklist and Annex K.

## EC 001.02 - ADVISE COMMAND ON MACHINERY/SYSTEM STATUS

**EC Time:** 15 mins (per trainee) + 15 min debrief (360 minutes)

**ASSESSMENT ACTIVITY**: Trainees will be given a scenario involving marine systems engineering ancillary/auxiliary component failure and/or environmental emergencies. Trainees will be required to identify the condition, formulate options using the EHM and provide recommendations. Trainee will advise command on options. The scenario will begin after IAs are completed by EWK.

**Pass/fail criteria**: 75% on all non-critical items. All critical items must be completed correctly.

**ASSESSMENT RESOURCES:**

- EHM;
- Scenario (various scenarios for each trainee);
- References;
- MCRT; and
- Assistance of EWK.

**IMPACT OF FAILURE:** Remedial Training / Re-test.

| Senior Engineering Watchkeeper – EC 001.02 Assessment Checklist | | | |
|---|---|---|---|
| **Candidate Name:** | **Assessor Name:** | **Date:** | **Session:** |

**INVIGILATOR NOTES:**

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 75% SATs on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.
3. Instructor act as senior engineering personnel/command.

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| **1. Duties and Responsibilities of Sr EWK** | | | |
| Check engineering log for the following: | | | |
| Ongoing maintenance; | | | |
| Defects; | | | |
| Running equipment; and | | | |
| Update the engineering log as necessary. | | | |
| Perform page rounds to include: | | | |
| Ancillary pages; | | | |
| Auxiliary pages; and | | | |
| Event viewer. | | | |
| Communicate to the MCR personnel and OOW* | | | |
| Assume control of the watch. | | | |
| **2. Operating in normal/abnormal system operation** | | | Begin scenario with a marine systems engineering problem from EO 001.02 |
| Ensure EWK conducts proper IA in accordance with A31/D2 to include: | | | |
| Responding to ancillary/auxiliary/environmental emergencies; and* | | | |
| Responding to ancillary/auxiliary alarms and faults or environmental emergency.* | | | |
| Conduct an initial situation brief to the officer of the watch to include:* | | | |
| Outlining the situation;* | | | |
| Impact; and* | | | |
| Options.* | | | |
| **3. Operating Equipment Health Monitoring (EHM)** | | | |
| Navigating EHM software to include: | | | |
| Identifying relevant ancillary/auxiliary signals; | | | |
| Selecting relevant ancillary/auxiliary signals; | | | |
| Formatting ancillary/auxiliary data; and | | | |
| Displaying ancillary/auxiliary data. | | | |

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| Analyzing EHM ancillary/auxiliary signals by: | | | |
| Comparing current data to identify issues and/or failures; and | | | |
| Compare data in event viewer to identify issues and/or failures of the event. | | | |
| **4. Advise command on marine systems capabilities and limitations** | | | |
| Report deficiencies to senior engineering personnel * | | | |
| Develop a plan considering limitations and restrictions based on scenario | | | |
| Update officer of the watch with recommendations to include: | | | |
| Update on the situation; | | | |
| Options; | | | |
| Impact; | | | |
| Estimated time of repair; and | | | |
| Brief is clear and concise. | | | |
| Brief the watch on the following: * | | | |
| Safety precautions;* | | | |
| Required repairs; and | | | |
| Operational hazards.* | | | |

| ADDITIONAL COMMENTS | |
|---|---|
| **STRENGTHS:** | **AREAS FOR DEVELOPMENT:** |

| THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT | | | |
|---|---|---|---|
| **OVERALL ASSESSMENT (circle one)** | | **FAIL** | **PASS** |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

**PO 001 - Advise command on marine systems operational capabilities and limitations**

**EO 001.03 - ADVISE COMMAND ON POWER GENERATION AND DISTRIBUTION SYSTEMS**

53. Performance. Advise command on Power Generation and Distribution systems.

54. Conditions.

Given:

(1) Assistance;

(2) References;

(3) Operational IPMS;

(4) EHM; and

(5) Supervision.

Denied. Nil.

Environmental:

(1) Classroom (DWAN); and

(2) Trainer (MCRT).

55. Standard. The trainee shall independently advise command on power generation and distribution systems (CT0030) IAW refs by:

Advising command on ships electrical load (CT0050);

Configuring ships electrical load to include (CT0060) to include:

(1) Decoking (MWM only);

(2) Action state; and

(3) Single switchboard/generator operation.

Reporting on power generation and distribution systems (CT0330) to include:

(1) High crankcase pressure;

(2) Blackout/brownout; and

(3) Full/partial ground faults.

56. Time Required:

EO – 800 mins to deliver training;

EC – 360 mins (total for all trainees); and

PC – 720 mins (total for all trainees).

57. Remarks. Nil.

58. Teaching Points.

| Teaching Points | Ref | Method / Media / Environment / Training Activity Details | Time (mins) |
|---|---|---|---|
| **LESSON OBJECTIVE: Configure ships electrical load** | | | |
| 1. Identify the 11 auto functions of the switchboard. | A15<br>A23 | **Method / Media**: IL.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture on the 11 auto functions of the switchboard.<br>**Trg Resources:** Lecture, handout. | 50min |
| 2. Conduct decoking (MWM only) to include:<br>a. Briefing command; and<br>b. Configuring electrical load to accommodate published decoking parameters. | A53 | **Method / Media**: IL.<br>**Trg Environment**: Classroom/MCRT.<br>**Activity Details:** Instructor will give a lecture on decoking and briefing command. Trainees may be given a scenario which requires a briefing to command on decoking procedures.<br>**Trg Resources:** Lecture, MCRT, scenario. | 200min |
| 3. Conducting single operation to include:<br>a. Switchboard:<br>i. Loss of redundancies to include ACOS/ MCOS/ SABT; and<br>ii. Loss of auxiliary machinery.<br>b. Generator:<br>i. Electrical load restrictions. | A15 | **Method / Media**: IL.<br>**Trg Environment**: Classroom/MCRT.<br>**Activity Details:** Instructor will give a lecture on conducting single operation and briefing command. Trainees may be given a scenario which requires a briefing to command on single switchboard/generation operation procedures.<br>**Trg Resources:** Lecture, MCRT, operational OBTS, scenario. | 200min |
| 4. Describing action state to include:<br>a. Expected results; and<br>b. Possible mitigation. | A15<br>A23<br>A31 | **Method / Media**: IL.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture on action state and briefing command.<br>**Trg Resources:** Lecture. | 50min |
| 5. Reacting to high crankcase pressure to include:<br>a. Ensure immediate actions are conducted:<br>i. Override/trip.<br>b. Mitigation options. | A15<br>A31<br>A40<br>A54 | **Method / Media**: IL.<br>**Trg Environment**: Classroom/MCRT.<br>**Activity Details:** Instructor will give a lecture on reacting to high crankcase pressure and briefing command. Trainees may be given a scenario which requires a briefing to command on immediate actions and mitigation procedures.<br>**Trg Resources:** Lecture, MCRT, scenario. | 200min |
| 6. Responding to unstable power situations to include:<br>a. Brownout:<br>i. Mitigation options. | A31 | **Method / Media**: IL.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture on responding to unstable power situations and briefing command. | |

| | | | |
|---|---|---|---|
| b. Blackout:<br>i. Ensure immediate actions are conducted; and<br>ii. Mitigation options. | | **Trg Resources:** Lecture. | |
| 7. Responding to full/partial ground faults to include:<br>a. Briefing command;<br><br>b. Briefing watch on safety considerations; and<br><br>c. Plant configuration considerations. | A31<br>A55 | **Method / Media**: IL.<br>**Trg Environment**: Classroom.<br>**Activity Details:** Instructor will give a lecture on responding to full/partial ground faults and briefing command.<br>**Trg Resources:** Lecture. | 50min |

59. <u>Test Details</u>:

a. EO 001.03 will be assessed by EC 001.03 and PC 001 IAW EC 001.03 Checklist and Annex K.

## EC 001.03 - ADVISE COMMAND ON POWER GENERATION AND DISTRIBUTION REQUIREMENTS

**EC Time:** 15 mins (per trainee) + 15 min debrief (360 minutes).

**ASSESSMENT ACTIVITY**: Trainees will be given a scenario involving a power generation distribution failure. Trainees will be required to identify the condition, formulate options using the EHM and provide recommendations. Trainee will advise command on options. The scenario will begin after IAs are completed by EWK.

**Pass/fail criteria**: 75% on all non-critical items. All critical items must be completed correctly.

**ASSESSMENT RESOURCES:**

- EHM, MCRT;
- Scenario (various scenarios for each trainee);
- References; and
- Assistance of EWK.

**IMPACT OF FAILURE:** Remedial Training / Re-test.

| Senior Engineering Watchkeeper – EC 001.03 Assessment Checklist | | | |
|---|---|---|---|
| **Candidate Name:** | **Assessor Name:** | **Date:** | **Session:** |

**INVIGILATOR NOTES:**

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 75% SATs on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.
3. Instructor act as senior engineering personnel/command.

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| **1. Duties and Responsibilities of Sr EWK** | | | |
| Check engineering log for the following: | | | |
| Ongoing maintenance; | | | |
| Defects; | | | |
| Running equipment; | | | |
| Update the engineering log as necessary. | | | |
| Perform page rounds of electrical systems and subsystems. | | | |
| Communicate to the MCR personnel and OOW* | | | |
| Assume control of the watch. | | | |
| **2. Operating PG&D in normal/abnormal modes** | | | Begin scenario with a PG&D problem from EO 001.03 |
| Ensure EWK conducts IA to include: | | | |
| Responding to PG&D emergencies; and* | | | |
| Responding to PG&D alarms and faults.* | | | |
| Conduct an initial situation brief to the officer of the watch to include: | | | |
| Outlining the situation;* | | | |
| Impact; and* | | | |
| Options.* | | | |
| **3. Operating Equipment Health Monitoring (EHM)** | | | |
| Navigating EHM software to include: | | | |
| Identifying relevant PG & D signals; | | | |
| Selecting relevant PG & D signals; | | | |
| Formatting PG & D data; and | | | |
| Displaying PG & D data. | | | |
| Analyzing EHM signals by: | | | |
| Comparing current data to identify issues and/or failures; and | | | |
| Comparing data in event viewer to identify issues and/or failures of the event. | | | |
| **4. Advise command on PG&D situation** | | | |

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| Report deficiencies to senior engineering personnel * | | | |
| Report deficiencies to senior engineering personnel * | | | |
| Develop a plan considering limitations and restrictions based on scenario. | | | |
| Update officer of the watch with recommendations to include: | | | |
| Update on the situation; | | | |
| Options; | | | |
| Impact; | | | |
| Estimated time of repair; | | | |
| Brief is clear and concise. | | | |
| Brief the watch on the following:* | | | |
| Safety precautions;* | | | |
| Required repairs; and | | | |
| Operational hazards.* | | | |

| ADDITIONAL COMMENTS | |
|---|---|
| **STRENGTHS:** | **AREAS FOR DEVELOPMENT:** |

| THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT | | | |
|---|---|---|---|
| **OVERALL ASSESSMENT (circle one)** | | **FAIL** | **PASS** |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

**PO 001 - Advise command on marine systems operational capabilities and limitations**

**EO 001.04 - ADVISE COMMAND ON BATTLE DAMAGE SURVEY RESULTS**

60. Performance. Advise command on battle damage survey results.

61. Conditions.

Given:

(1) Assistance;

(2) References;

(3) Operational IPMS; and

(4) Supervision.

Denied. Nil.

Environmental:

(1) Classroom (DWAN); and

(2) Trainer (MCRT).

62. Standard. The trainee shall independently advise command on battle damage survey results (IT0010) IAW refs by:

Reporting on battle damage due to multiple equipment failures caused by:

(1) Fire:

(a) Conducting immediate actions;

(b) Assessing battle damage to engineering systems (IT0120);

(c) Identifying impact on operations; and

(d) Considering mitigations to include:

(i) Tactical ventilation;

(ii) Boundaries into engineering spaces;

(iii) Power isolation;

(iv) Engineering equipment changes; and

(v) Provide recommendations.

(2) Flood:

(a) Conducting immediate actions;

(b) Assessing battle damage to engineering systems (IT0120);

(c) Identifying impact on operations;

(d) Considering mitigations to include:

(i) Boundaries into engineering spaces;

(ii) Power isolation;

(iii) Consider impact on stability; and

(iv) Engineering equipment changes.

(e) Providing recommendations.

(3) Grounding/Collision:

Conducting immediate actions;

Assessing battle damage to engineering systems (IT0120);

Identifying impact on operations;

Considering mitigations to include:

(i) Boundaries into engineering spaces;

(ii) Power isolation;

(iii) Consider impact on stability;

(iv) Engineering equipment changes; and

(v) Provide recommendations.

(4) Obeying telegraphs by:

(a) Reacting to engineering emergencies/action stations.

63. Time Required:

EO – 1250 mins to deliver training;

EC – 360 mins (total for all trainees); and

PC – 720 mins (total for all trainees).

64. Remarks. Nil.

65. Teaching Points.

| Teaching Points | Ref | Method / Media / Environment / Training Activity Details | Time (mins) |
|---|---|---|---|
| **LESSON OBJECTIVE: Reporting on battle damage due to multiple equipment failures/fire** | | | |
| 1. Responding to fire in engineering space to include:<br>a. Conducting immediate actions:<br>i. Assessing battle damage to engineering systems;<br>ii. Identify impact on operations;<br>iii. Consider mitigations to include:<br>a) Tactical ventilation;<br>b) Boundaries into engineering spaces;<br>c) Power isolation; and | A10<br>A30<br>A31<br>A46<br>A47<br>B1<br>D1<br>D2 | **Method / Media**: IL/scenario.<br>**Trg Environment**: Classroom/MCRT.<br>**Activity Details:** Instructor will give a lecture on reacting to fire in an engineering space and briefing command. Trainees may be given a scenario which requires a briefing to command on immediate actions and mitigation procedures.<br>**Trg Resources:** Lecture, MCRT, scenario. | 400min |

| Teaching Points | Ref | Method / Media / Environment / Training Activity Details | Time (mins) |
|---|---|---|---|
| d) Engineering equipment changes.<br>iv. Providing recommendations. | | | |
| **LESSON OBJECTIVE:** Reporting on battle damage due to multiple equipment failures/flood | | | |
| 1. Responding to flood in engineering space to include:<br>a. Conducting immediate actions to include:<br>i. Assessing battle damage to engineering systems;<br>ii. Identifying impact on operations; and<br>iii. Considering mitigations to include:<br>a) Boundaries into engineering spaces;<br>b) Power isolation;<br>c) Impact on stability; and<br>d) Engineering equipment changes.<br>iv. Provide recommendations. | A10<br>A30<br>A31<br>A45<br>A46<br>A47<br>B1<br>D1<br>D2 | **Method / Media**: IL/scenario.<br>**Trg Environment**: Classroom/MCRT.<br>**Activity Details:** Instructor will give a lecture on reacting to flood in an engineering space and briefing command. Trainees may be given a scenario which requires a briefing to command on immediate actions and mitigation procedures.<br>**Trg Resources:** Lecture, MCRT, scenario. | 400min |
| **LESSON OBJECTIVE: Reporting on battle damage due to multiple equipment failures/collisions and grounding** | | | |
| 1. Responding to grounding/collision in engineering space to include:<br>a. Conducting immediate actions to include:<br>i. Assessing battle damage to engineering systems;<br>ii. Identifying impact on operations; and<br>iii. Considering mitigations to include:<br>a) Boundaries into engineering spaces;<br>b) Power isolation;<br>c) Impact on stability; and<br>d) Engineering equipment change.<br>iv. Provide recommendations. | A10<br>A30<br>A31<br>A45<br>A46<br>A47<br>B1<br>D1<br>D2 | **Method / Media**: IL/scenario.<br>**Trg Environment**: Classroom/MCRT.<br>**Activity Details:** Instructor will give a lecture on reacting to collision and/or grounding that effects an engineering space and briefing command. Trainees may be given a scenario which requires a briefing to command on immediate actions and mitigation procedures.<br>**Trg Resources:** Lecture, MCRT, scenario. | 400min |
| **LESSON OBJECTIVE: Complying to command order to obey telegraphs** | | | |
| 1. Reacting to engineering emergencies/action stations. | A31 | **Method / Media**: IL/scenario.<br>**Trg Environment**: Classroom/MCRT.<br>**Activity Details:** Instructor will give a lecture and brief command in responding to the executive order to obey telegraphs. Trainees may be given a scenario which requires a | 50min |

| Teaching Points | Ref | Method / Media / Environment / Training Activity Details | Time (mins) |
|---|---|---|---|
| | | briefing to command on immediate actions and mitigation procedures.<br>**Trg Resources:** Lecture, MCRT, scenario. | |

66. Test Details.

a. EO 001.04 will be assessed by EC 001.04 and PC 001 IAW EC 001.04 Checklist and Annex K.

## EC 001.04 - ADVISE COMMAND ON BATTLE DAMAGE SURVEY RESULTS

**EC Time:** 15 mins (per trainee) + 15 min debrief (360 minutes).

**ASSESSMENT ACTIVITY**: Trainees will be given a scenario involving battle damage with multiple equipment failures. Trainees will be required to identify the condition, formulate options and provide recommendations to command. At any time command may give the executive order to obey telegraphs.

**Pass/fail criteria**: 75% on all non-critical items. All critical items must be completed correctly.

**ASSESSMENT RESOURCES:**

- Scenario (various scenarios for each trainee);
- References;
- MCRT; and
- Assistance of EWK.

**IMPACT OF FAILURE:** Remedial Training / Re-test.

| Senior Engineering Watchkeeper – EC 001.04 Assessment Checklist | | | |
|---|---|---|---|
| **Candidate Name:** | **Assessor Name:** | **Date:** | **Session:** |

**INVIGILATOR NOTES:**

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 75% SATs on the remainder of the checklist items to be successful with this assessment.
2. Trainee will respond to executive command to obey telegraphs if ordered by invigilator/instructor.
3. IMPACT OF FAILURE: Remedial training, re-test.
4. Instructor act as senior engineering personnel/command.

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| **1. Duties and Responsibilities of Sr. EWK** | | | |
| Check engineering log for the following: | | | |
| Ongoing maintenance; | | | |
| Defects; | | | |
| Running equipment; and | | | |
| Update the engineering log as necessary. | | | |
| Perform page rounds of electrical systems and subsystems. | | | |
| Communicate to the MCR personnel and OOW* | | | |
| Assume control of the watch. | | | |
| **2. Responding to battle damage** | | | Begin scenario with a DC problem with multiple system failures from EO 001.04 |
| Conducting immediate actions to include:* | | | |
| Assessing battle damage to engineering systems; and* | | | |
| Identifying impact on operations.* | | | |
| Considering mitigations to include: | | | |
| Boundaries into engineering spaces;* | | | |
| Power isolation;* | | | |
| Impact on stability(flood, collision, grounding)/ Tactical ventilation (fire); and* | | | |
| Engineering equipment changes. | | | |
| Conduct an initial situation brief to the officer of the watch to include: * | | | |
| Outlining the situation;* | | | |
| Impact; and* | | | |
| Options.* | | | |
| Responding to executive order to obey telegraphs (IAW scenario) * | | | |
| **3. Assess battle damage to engineering systems** | | | |

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| Report deficiencies to senior engineering personnel. | | | |
| Develop a plan which considers limitations or restrictions based on scenario. | | | |
| Update officer of the watch with recommendations to include: | | | |
| Update on the situation; | | | |
| Options; | | | |
| Impact; | | | |
| Estimated time of repair; and | | | |
| Brief is clear and concise. | | | |
| Ensure the watch is briefed on the following: | | | |
| Safety precautions;* | | | |
| Required repairs; and* | | | |
| Operational hazards.* | | | |

| ADDITIONAL COMMENTS | |
|---|---|
| **STRENGTHS:** | **AREAS FOR DEVELOPMENT:** |

| THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT | | | |
|---|---|---|---|
| **OVERALL ASSESSMENT (circle one)** | | **FAIL** | **PASS** |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

## PC 001 - ADVISE COMMAND ON MARINE SYSTEMS OPERATIONAL CAPABILITIES AND LIMITATIONS

**PC Time:** 720 mins (45 min/per trainee) + 15 min debrief.

**ASSESSMENT ACTIVITY**: Trainee will be given a scenario that includes sailing through emergency and battle damage events; and operating the plant and engineering machinery outside of design intent. Scenario shall include three of the following problems with: main propulsion, ancillaries, auxiliaries, power generation and distribution (PG&D) and/or battle damage.

**Pass/fail criteria**: All critical items must be complete. 75% of the non-critical items must be complete to pass.

**ASSESSMENT RESOURCES:**

- IPMS console;
- Scenarios;
- References; and
- Real or simulated rounds person.

**IMPACT OF FAILURE:** Remedial Training / Re-test / TRB after second unsuccessful attempt.

| Senior Engineering Watchkeeper – PC 001 Assessment Checklist | | | |
|---|---|---|---|
| **Candidate Name:** | **Assessor Name:** | **Date:** | **Session:** |

**INVIGILATOR NOTES:**

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]) and also achieve a 75% SATs on the remainder of the checklist items to be successful with this assessment.
2. Trainee will respond to executive command to obey telegraphs when ordered by invigilator/instructor.
3. IMPACT OF FAILURE: Remedial training, one re-test, and TRB is unsuccessful on the re-test.
4. Instructor act as senior engineering personnel/command.

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| **1. Duties and Responsibilities of Sr EWK** | | | |
| Check engineering log for the following: | | | |
| Ongoing maintenance; | | | |
| Defects; | | | |
| Running equipment; and | | | |
| Update the engineering log as necessary. | | | |
| Perform page rounds of electrical systems and subsystems. | | | |
| Communicate to the MCR personnel and OOW.* | | | |
| Assume control of the watch. | | | |
| **2. Responding to battle damage** | | | Begin scenario with a DC problem with multiple system failures from PO 001 |
| Conducting immediate actions to include: | | | |
| Assessing battle damage to engineering systems; and* | | | |
| Identifying impact on operations.* | | | |
| Considering mitigations to include:* | | | |
| Boundaries into engineering spaces;* | | | |
| Power isolation;* | | | |
| Impact on stability (flood, collision, grounding)/ Tactical ventilation (fire); and* | | | |
| Engineering equipment changes.* | | | |
| Conduct an initial situation brief to the officer of the watch: | | | |
| Outlining the situation;* | | | |
| Impact;* | | | |
| Options; and* | | | |
| Responding to executive order to obey telegraphs.* | | | |
| **3. Assess battle damage to engineering systems** | | | |

| ITEM | SAT | UNSAT | COMMENTS |
|---|---|---|---|
| Report deficiencies to senior engineering personnel.* | | | |
| Develop a plan which considers limitations and restrictions based on scenario. | | | |
| Update officer of the watch with recommendations to include: | | | |
| Update on the situation; | | | |
| Options; | | | |
| Impact; | | | |
| Estimated time of repair; and | | | |
| Brief is clear and concise. | | | |
| Ensure the watch is briefed on the following: | | | |
| Safety precautions;* | | | |
| Required repairs; and | | | |
| Operational hazards.* | | | |

| ADDITIONAL COMMENTS | |
|---|---|
| **STRENGTHS:** | **AREAS FOR DEVELOPMENT:** |

| THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT | | | |
|---|---|---|---|
| **OVERALL ASSESSMENT (circle one)** | | **FAIL** | **PASS** |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

# ANNEX A – MAIN REFERENCES

A - CF or other official Canadian government publication.

B - Official Allied Force or Allied government publication.

C - Commercial publication.

D - Other.

| Refs Code | Publication Number | Title |
|---|---|---|
| A1 | C-25-512-000/MS-001 | Auxiliary Seawater Circulating System |
| A2 | C-23-HFX-000/MB-001 | HALIFAX Class Propulsion Operating Manual |
| A3 | C-27-784-000/MS-001 | Low Pressure Compressed Air System Manual |
| A4 | C-27-783-000/MS-001 | High Pressure Compressed Air System Manual |
| A5 | C-24-541-000/ME-001 | Engine Instruction LM2500 Gas Turbine System |
| A6 | C-24-541-000/MF-001 | Schedule Maintenance Engine LM2500 Gas Turbine System |
| A7 | C-27-302-000/MS-001 | Main Propulsion Gearing Manual (Vol. 1, Book 1) |
| A8 | C-25-510-000/MS-001 | Fuel Oil Filling System Manual |
| A9 | C-24-293-000/MS-001 | Main Propulsion System Manual |
| A10 | C-23-HFX-000/MS-002 | Canadian Navy Damage Control Information Book for CPF |
| A11 | C-24-547-000/MS-002 | Hydraulic Start System for CPF 330 to 335 |
| A12 | C-24-547-000/MS-001 | Hydraulic Start System for CPF 336 to 341 |
| A13 | C-25-509-000/MS-001 | Fuel Oil Service System |
| A14 | C-25-550-000/MS-001 | Fuel Filter Water Separator |
| A15 | C-26-320-000/MS-001 | Primary Electrical Power Generation and Distribution |
| A16 | C-41-541-000/MM-001 | LM2500 Gas Turbine System |
| A17 | C-24-545-A00-NY-001 | Technical Manual, Main Shaft Flexible Coupling |
| A18 | C-24-545-B00/MS-001 | Technical Manual, Main Thrust Block Bearing |
| A19 | C-24-545-000/MS-001 | Stern Shaft Seal and Bulkhead Seal |
| A20 | C-24-545-D00/MS-001 | Technical Manual, Plummer Bearing |
| A21 | TBD-Drafts | IPMS for HALIFAX Class Equipment (Description) |
| A22 | TBD-Drafts | IPMS for HALIFAX Class Equipment (Data Summaries) |
| A23 | TBD-Drafts | IPMS Operating Instructions Manual |
| A24 | C-29-345-000/MS-001 | System Manual, Main Refrigeration |
| A25 | C-27-785-000/MS-001 | Fresh Water System |
| A26 | C-27-782-000/NQ-001 | Oil Pollution Abatement and Bilge Stripping System |
| A27 | C-27-790-000/MS-001 | Sanitary System Manual |
| A28 | C-27-867-000/MS-001 | Service Steam System |
| A29 | C-29-355-000/MS-001 | Heating, Ventilation and Air Conditioning Manual, Books 1 , 2 |

| | | |
|---|---|---|
| A30 | C-03-005-033/AA-000 | Naval Engineering Manual (NEM), Vol. 1 |
| A31 | C-03-050-HFX/MB-002 | Manual of Emergency Procedures Marine Systems Engineering (at sea) |
| A32 | C-25-539-000/MS-001 | Motor Driven Lube Oil Pump Technical Manual |
| A33 | C-03-010-578/MS-F01 | CPF 690222 Moisture Separator System (LM2500) |
| A34 | C-25-547-000/MS-001 | OSDA 7-02-366 Lube Oil Separator |
| A35 | C-24-559-000/MS-001 | Controllable Pitch Propeller and Propeller Shafting |
| A36 | C-24-548-000/MS-001 | 20 PA6 V 280 Pielstick Technical Manual, Vol. 1 |
| A37 | C-24-559-000/TB-001 | Technical Bulletin Marine CRPP Control Oil Cooler Operating Instruction – Applicable to HALIFAX Class Vessel |
| A38 | C-27-778-000/MS-001 | Operating and Maintenance Instruction with parts Identification List Steering Gear Control System |
| A39 | C-27-945-000/MS-001 | Operation and Maintenance Instruction with Parts Identification List Hydraulic System (Wagner) Steering Gear Control System |
| A40 | C-26-321-000/MS-001 | Generator Set Diesel 850KW Vol 1 |
| A41 | C-26-321-000/MS-002 | Generator Set Diesel 850KW Vol 2 |
| A42 | C-27-791-000/MS-001 | Sea Water Services |
| A43 | C-24-555-000/MS-001 | Technical Manual for Gas Turbine Water Wash Tank |
| A44 | C-26-411-000/TB-001 | MWM 602 Technical Bulletin Decoking Instructions |
| A45 | C-23-HFX-000/MS-003 | Manual of Trim and Stability Applicable to Halifax Class |
| A46 | N/A | Ship’s Standing Orders AL11 |
| A47 | N/A | CFCD 132 –DC Manual |
| A48 | C-03-005-012/AM-001 | Naval Material Management Systems (NaMMS) |
| A49 | C-24-548-000/TB-001 | Propulsion Cruise Diesel Engine PIELSTICK 20 PA6 V 280 MPC |
| A50 | C-82-010-010/TB-001 | Microbiological Growth in Shipboard Fuel |
| A51 | C-82-010-011/TB-003 | Lubrication of HP/LP Compressors |
| A52 | C27-886-000/TB-001 | Low Pressure Compressed Air Compressor Rotor Stator Unit |
| A55 | D-03-003-005/SF-000 | General Electrical Specification For Canadian Forces Ships |
| A56 | B-GG-380-000/FP-002 | Diving Safety Checklist |
| A57 | C-23-HFX-000/TB-001 | Sea Water Ballast/Fuel Tank Usage |
| A58 | C-23-HFX-000/TB-003 | LM2500 Post Shutdown Fan |
| A59 | C-24-010-020/TB-001 | Corrosion Control Diesel |
| A60 | C-24-559-000/TB-001 | CRPP Cooler Operating Instructions |
| A61 | C-24-637-000/TB-001 | Cleaning GT Demisters |
| A62 | C-24-559-000/TB-002 | CRPP Oil Contamination |
| A63 | C-39-143-000/TB-001 | HIFR Hose |
| A64 | C-24-020-001/TB-001 | Diesel Engine Lube Oil Testing |

| | | |
|---|---|---|
| A65 | DMSS 3401 | Fixed Pitch Operation |
| A66 | Technical Instruction | Main Lube Oil Operation and Maintenance |
| B1 | B-GN-007-RCN/FP-001 | CFCD 132-RCN DC Manual |
| C1 | N00023-80-C2080 | Instruction for MLO Relief Valve |
| D1 | N/A | Ship specific EOTIs |
| D2 | N/A | Halifax Class Safety Environmental Management System |
| D3 | N/A | Captain’s Ship Book |

# ANNEX B – TASK LIST

| Task No | Task Statement | Trg / No Trg | PO |
|---|---|---|---|
| CT0020 | Advise command on marine systems operational capabilities and limitations | Trg | 001 |
| CT0040 | Advise command on propulsion capabilities and limitations | Trg | 001 |
| CT0010 | Advise command on marine systems engineering parameters | Trg | 001 |
| CT0030 | Advise command on power generation and distribution systems | Trg | 001 |
| CT0050 | Advise command on ship's electrical load | Trg | 001 |
| CT0060 | Configure ship's electrical load | Trg | 001 |
| CT0310 | Report on machinery limitations | Trg | 001 |
| CT0320 | Report on marine systems engineering parameters | Trg | 001 |
| CT0330 | Report on power generation and distribution systems | Trg | 001 |
| CT0340 | Report on propulsion deficiencies | Trg | 001 |
| CT0220 | Manage engineering pre-sailing checks (event) | No Trg | NA |
| GT0010 | Advise command on machinery/system status | Trg | 001 |
| HT0070 | Operate IPMS EHM | Trg | 001 |
| IT0010 | Advise command on battle damage survey results | Trg | 001 |
| IT0040 | Advise command on damage control system status | No Trg | NA |
| IT0120 | Assess battle damage to engineering systems | Trg | 001 |
| OT0210 | Supervise liquid cargo transfer (SEMS) | No Trg | NA |
| OT0010 | Advise Command on Fuel Quality | No Trg | NA |
| WT0030 | Analyze ship’s stability | No Trg | NA |
| WT0050 | Investigate ship’s stability | No Trg | NA |

# ANNEX C QSP WRITING BOARD RECORD OF PROCEEDINGS

Dates: 25 Jun-11 Jul 18

Location: WP 1020 room 105C

Board Members:

| Position | Rank and name | MOSID / Unit | Dates |
|---|---|---|---|
| TM | T70 698 960 CPO2 Gagne | NTDC(P) | 25 Jun-11 Jul 18 |
| Chair | T42 506 160 CPO2 Cox | NTDC(P) | 25 Jun-11 Jul 18 |
| SME | C15 166 180 CPO2 Zsombor | NTDC(P) | 25 Jun-11 Jul 18 |
| SME | V82 541 093 PO1 Sleeman | NFS(P) | 25 Jun-11 Jul 18 |
| SME | B11 813 554 PO1 Penney | NTDC(P) | 25 Jun-11 Jul 18 |
| SME | T14 370 479 PO1 Albert | FDU(A) | 25 Jun-11 Jul 18 |
| SME | F35 572 652 MS Vallee | NTDC(P) | 25 Jun-11 Jul 18 |
| TDO | F11 652 350 SLt Hamilton<br>A64 351 534 Lt(N) Robinson | NTDC(P) | 25 Jun-11 Jul 18 |
| Secretary | A64 351 534 Lt(N) Robinson | NTDC(P) | 25 Jun-11 Jul 18 |

Board Objective: To develop the QSP for the MAR TECH SNR Engineering Watchkeeper USQ in the overall context of the Marine Technician.

Discussions / Decisions:

| Decision/Event | Discussion / Detail |
|---|---|
| **Day 1: 25 Jun 18** | |
| Intro/Train/No Train | Board commenced with TDO delivering introduction brief. Board reviewed the training assessment conducted by a previous board to confirm the concept of the Sr EWK's role. The SMEs then reviewed the trained/no trained tasks from the training assessment to ensure they captured all of the required tasks. It was determined that *CT 0220 Manage Engineering Pre-Sail Checks (event)* needed to have a flash-up component added. |
| Task Grouping | The board commenced task groupings under guidance from the TDO. Board completed initial groupings and then determined additional steps required to complete the process. |
| TPA | Conducted TPA with Board. Characteristics of the learner are captured in Annex M. |
| **Day 2: 26 Jun 18** | |
| Task Refining | The board continued with defining the steps required for the Sr EWK on the job. The board re-examined the topic of battle damage related tasks. It was determined that task *IT0040 Advise command on damage control system status* would be a no train due to the task not being in the scope of the Sr EWK job. After deliberation the tasks appeared to all fall under one Performance Objective. *Task CT0220 Manage engineering pre-sailing checks (events)* was determined to be no train due to experience. |

| Decision/Event | Discussion / Detail |
|---|---|
| Drafting POs | The board conducted a discussion regarding the exact scope covered by *Advise command on machinery/system status (GT0010)* and *Advise command on marine systems operational capabilities and limitations (CT0020).* It was determined that the two tasks would be switched as *Advise command on marine systems operational capabilities and limitations (CT0020)* was felt to be more overarching title for the Performance Objective. |
| Format of trg | There was some debate as to where the PC should take place in order to replicate training as closely as possible. There was discussion of conducting the PC on ship, in a trainer at DC Div, as a board at the school. The issue was put in the parking lot to continue with development of the QSP. The board will revisit the issue later. |
| **Day 3: 27 Jun 18** | |
| Format of trg | The board discussed the different options as well as the scope of training. There was a decision to seek information from the TNA emails to confirm the way ahead for this course. The board feels they are not in a position to determine delivery method until training has been further defined. |
| Developing EOs and Learning Objectives | The board moved into developing EOs beginning with Advising Comd on propulsion. They began with identifying the activity to train this EO. It was determined there would be learning objectives within the EO that would necessitate activities as well. This will lead to a group of activities developed to train the EO. One board mbr indicated that it's important to train the process at the outset so that it can then be applied to all EOs. |
| Brainstorm Tasks For Propulsion Caps and Lims | The board brainstormed skills and knowledge needed in support of the propulsion EO. The list of items has been deemed imperative for Sr EngrWK. They are expected to have a higher level of knowledge related to the operation and follow on actions when operating outside normal parameters.<br><br>This list of requirements delivery: could be independent study package to be completed before the crse (during their 500/180/12 as part of WK) or at the beginning of the course. **Trg Mgr:** Discussion with TM is that this will be integrated into the course rather than a PIP. In future it could become DL. Nonetheless, the content doesn't change, just how it is delivered will change. |
| Sr Engr Watchkeeper AOR question to Trg Mgr | **What is the Sr Engr WK doing when normal steaming is occurring**? **Are they in the MCR at all times?** During abnormal operation, are they expected to manage IPMS only or are they leading maintenance repairs during emergency stations? How many qual Sr Engr WK will be on board at any given time?<br>**Trg Mgr:** Initial intent is the Sr Eng WK will remain in the MCR but will liaise with Occ Mgr. MTF The Senior Watchkeeper will have to make himself available while on Watch. They will ensure he can be reached via SHINCOM and exact location.<br>In case where an emergency would occur, the Eng Watchkeeper is to make the plant safe and then call the Snr Watchkeeper immediately. |
| Sr Engr Watchkeeper Decision Making Abilities | The Sr Watchkeeper needs to be well versed in safety concerns for personnel, eqpt and the platform overall. This involves a level of risk management and being able to make critical decisions quickly and be confident to make a call and follow through with those decisions. |
| EOTI | Engineering Officers Technical Instructions to be added to Abbreviations |

| **Day 4: 28 Jun 18** | |
|---|---|
| Thermostatic release valve | The board continued to design EO 001.01.CPO2 Zsombor was ill today, in his stead CPO2 Drew participated in the Board. In EO 001.01 thermostatic control valve was seen by the board to refer to the Walton valve currently in use by the fleet. For the use of the references in the para 6 of the EO portion of the QSP, the Board felt that due to the possibility of rapid changes to some of the references that they would not proceed into the minutia of references. |

| | |
|---|---|
| IPMS refs | The EHM LO are missing the IPMS references that are still to be promulgated. Training will be provided by SMEs until manuals are available. |
| Trainer time | The EWK and Sr EWK can be run concurrently to maximize trainer availability |
| | The board completed EO 001 including the EC not including the EC checklist |
| **Day 5: 03 Jul 18** | |
| | Board convened after the weekend, CPO2 Zsombor is not returning to the board. SLt Horvat has assumed the job of board secretary. |
| EHM not critical<br>Pass criteria | Board determined that the EHM initial EC in EO001 is not critical<br>Develop an effective plan determined as non-critical due to engineering watch present and advising senior EWK on situation<br>Remainder of non-critical items; Passing grade: 75% |
| Knowledge threshold Cert 3 | Current state of the knowledge threshold for cert 3 is above the requirement of the senior EWK. The board determined that not all of the material that a cert 3 is required to learn will be required of the Sr EWK. |
| N/T CT0010 | Board included some fundamental knowledge regarding CT0010 but determined that it was already trained and that those subjects will be covered in abnormal conditions like EO001.01 |
| PLQ prerequisite | Board chair reminded us that PLQ should be a qualification for this position, not just acting/lacking master seaman. This prerequisite will be added to the QSP document |
| | Board adjourned at 1445 after completing EO 001.02 EC checklist. |
| **Day 6: 04 Jul 18** | |
| | Board convened at 0800 and will begin with EO 001.03 Advise command on power generation and distribution systems. |
| CAT –decoking<br><br>Cmd approval-<br>U/S equip out of stby | Board is inquiring regarding if the new generators (CAT) require decoking. Board debated/questioned where power isolation should be taught (damage control EO). Board questioned if Sr EWK required to know when command approval is required prior to take standby/unserviceable equipment out of standby. This content to be incorporated in the training (Duties and responsibilities or in a specific subject area). |
| S EWK-<br>doecoking | PO2 Gillies advised the board that the S EWK should take over until electrician arrives on scene regarding decoking. |
| CAT-decoking | Board determined through conversation with LCMM (Bruce Blodgett) CATs engines do not require decoking therefore decoking comments only occurs on MWM engines. |
| S EWK responsibilities – DC events | Board discussed where the DC organization responsibility lies.<br>Came to the conclusion the DC may best fit into duties and responsibilities, as the EWK is already responsible for bringing ship into emergency stations.<br>HQ1 consults with XO during action station/survival stations.<br>Board agreed MCR's job is to assess battle damage results.<br>Board discussed OJT as an option for emergency response training (wartime) and TDO suggested to incorporate some of the DC elements into the prior EO's. The OJT for this phase would be more appropriate (minimal guidance).<br>Based on the past 3 EO's, the student should have been taught everything and the ship would cover on the job experience and that much of the realistic training would occur on the ship regarding DC after battle damage.<br>CPO2 Cox will consult with D NAV P&T in Ottawa regarding how to tackle the DC portion of the EO.<br>After some brainstorming and potential DC ideas, board is going to seek further clarification and reconvene |
| | Discussion about Sr EWK during battle damage scenario:<br>CPO2 Gagne said the roundsperson, who reports to Sr EWK do surveys on machinery and section based team reports to HQ1.<br>Sr EWK takes the report to officer of the watch and at that point the chief takes it over.<br>After the IA the sr EWK has no further responsibilities as it has gone up to chief.<br>Therefore may look at it pre IA. Sr EWK reports battle damage, assess initial impact, makes recommendations and briefs chief. |

| | |
|---|---|
| S EWK role starts pre IA for EO 001.04<br><br>Questions for CPO1 Pryor | Chief would come to the senior staff and asks to pass the information on<br>P1/Chief is the link between the Sr EWK, sr mechanical, and sr electrician and not advising anything as they are busy during action stations on their console<br>Sr EWK has nothing to do with HQ1 and will remain during action station, however would be the first person to identify initial damage, recommendations, and reports to the EO, afterwards, damage control takes over.<br><br>PRE IA=YES an enabling objective (identify damage to his engineering systems)<br>POST IA=NO not a enabling objective (can't advise command)<br><br>CPO2 Gagne and CPO2 Cox will contact Ottawa D NAV P&T with phone call and email CPO1 Pryor (6:30am and 7:00am)<br>Can add as teaching points; brief AT to ER spaces, tactical ventilation, IA of action stations, tactical consolidation of liquids, emergencies, abandon ship<br><br>OJT remains on the table: doing scenarios at the consoles and drills after this course?<br>How can you test/confirm/signatures/PO check/board/board scenario? |
| **Day 7: 05 Jul 18** | |
| | Board commences at 0800 and revisited discussions as per July 4 |
| OJT/Board answers from CPO1 Pryor | As per CPO2 Cox conversation with CPO1 Pryor, it was determined that Sr EWK will handle multiple system damages concurrently if the EO/CH ENG proceed to the scene of the damage.<br><br>OJT is to be minimized and used only when necessary to confirm training.<br><br>CPO1 Pryor agreed that a board may be included however, NTDC would create a standardized set of questions to conduct a board scenario if required.<br>Direction was to proceed with EO 001.04, damage control; fire, flood, condition, and grounding. EO 001.04 must be PRE IA of sr EWK prior to damage control taking over.<br>Sr EWK scenarios to consist of : IA, assess damage, report to command, advise command on solutions, obey telegraphs. |
| PC criteria | Board recommends a maximum of 3 engineering machinery failures for the performance check<br>Will possibly suggest recommended scenarios for development that are manageable (minor/major issues) |
| OJT | Board discussed including a board as a portion of the OJT and ultimately decided to revisit the issue |
| **Day 8: 06 Jul 18** | |
| | Board convened at 0800. |
| Scenario development NTDC (P) | Board determined that the scenarios will be developed by NTDC (P) to include abnormal/battle situations. Recommend that developers consult Sea Training Mar Tech personnel to develop scenarios. |
| Discussion of oral board<br><br>Scope of oral board | Board discussed the pros and cons of adding a board to the OJT package. The board is seen as a verification to the Sr EWK ability to answer complex questions to confirm knowledge, confidence. Board debated whether or not there is enough material to justify a board. Questions on a board would be focused on a new perspective as Sr EWK which is new material. Board determined that there was a requirement to conduct a board to have a defined standard to confirm knowledge.<br><br>Board recommended that the scope of the board would be focused on three questions to include: a system drawing (which will be provided) description limited to location, |

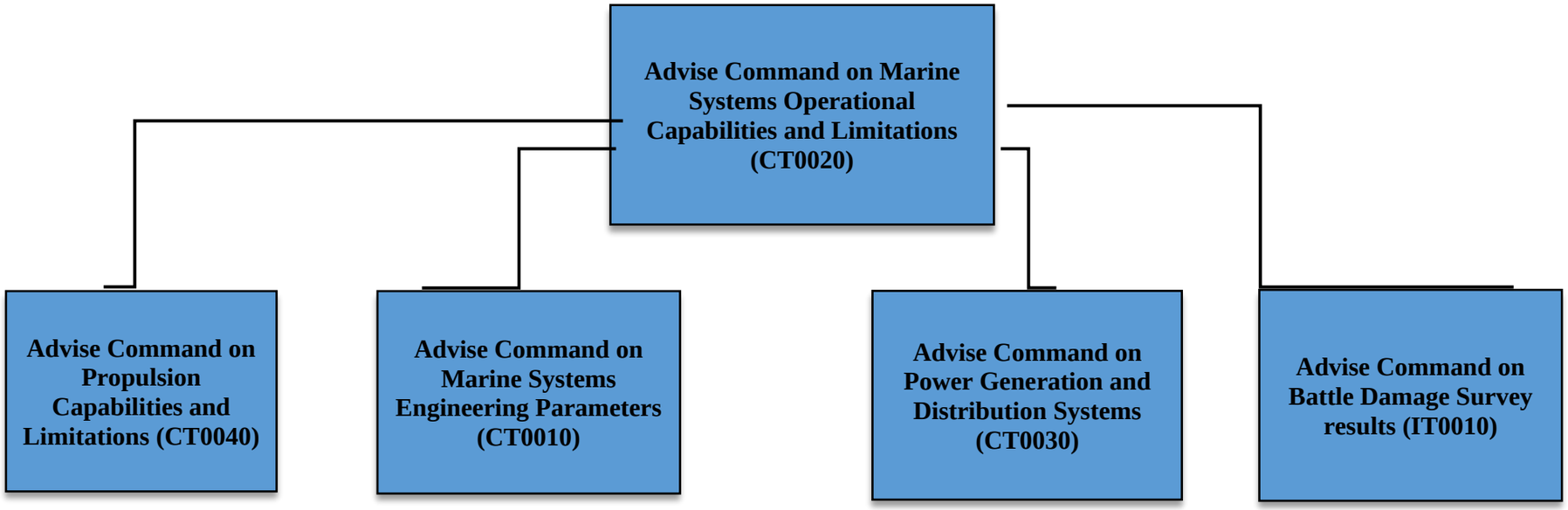
| | |
|---|---|
| | purpose and parameters, an abnormal/battle drill scenario, Operation Honour questions with EO and CHOD, Sr Mechanical Mar Tech. |
| Oral Board recommendation | Board recommended a board be added to the OJPR for the following reasons: To confirm that the trainee possesses the required consolidated knowledge of basic marine systems, advising command in abnormal operating conditions, and operation honour. To safely operate as a Sr EWK on a Halifax class ship. |
| | Board completed the Chapter 3 documents |
| | Board began work on completing Annexes and compiling the QSP document |
| **Day 9: 09 Jul 18** | |
| | Board convened at 0800 and continued to complete Annexes |
| | Board completed final review of ch1-3 and Annexes |
| Discussion of Oral Board<br><br>Operation Honour training | Board revisited the requirement for a board to complete the OJPR. LCdr Day asked the board to justify the need for a board to determine the distinction between the Sr EWK and the EWK board. The board identified some alternatives to include: cascading drills, a system walkthrough. In order to replicate training closely there may not be a requirement. There is a movement away from boards when developing training overall.<br><br>The direction from CO NTDC(P) is that the requirement for Operation Honour may not exist once the formal training is in place. (TBD within the next 6 months) Board discussed the requirement for a board if personnel are to conform to the Transport Canada requirements. The board discussed that a board is a final confirmation of the Sr EWK ability. To align with IMO International Maritime Organization. |
| **Day 10:10 Jul 18** | |
| | Board convened at 0800. |
| Oral Board Discussion | Board chair discussed the chances of a board not being accepted. He also discussed the culture change within the training system that is moving away from oral boards. B/C suggested a system walkthrough that is geared toward operators. |
| | A board will confirm that a Sr EWK can make critical decisions independently, without assistance of others in the MCR. This may not be possible in a drill scenario.The Sr EWK needs to be able to stand on his own in a battle situation/abnormal operations if the CH ENG is unavailable. Standard statements require Sr EWK to function independently |
| | A one on one assessment will take member out of his comfort zone. A quiet environment without distractions or assistance. |
| | PO1 Penney was absent due to and appointment. MS Vallee was absent due to a dental appt. (in the afternoon) PO1 Sleeman was attending an IPMS meeting but returned to discuss the new second COA and he was in agreement. |
| | The board discussed an option of a system walkthrough to replace the oral board. The board attempted to create a standard for the system walkthrough. |
| New option to replace oral board | Board members discussed and created another option to present as an alternative to the oral board. The practical assessment to include questions to assess critical thinking and advising command and can be included as requirements in the OJPR. The board chair will recommend and provide rationale for keeping a board with the OJPR. The alternative is to be presented if the oral board is not accepted. |
| | The board adjourned at 1500 |
| **Day 11, 11 Jul 18** | The board convened at 0800. |

| | |
|---|---|
| Oral Board discussion | Board members decided that the second option to replace a board would be acceptable if the preferred option of keeping an oral board is not accepted. |
| | Board adjourned at 0930 after reviewing the final document. |
| **28 Jul 20** | After discussions with the Div Cdr it was reinforced that this Qualification must be completed at sea. To ensure that the QSP was corrected the line" If a sea-going unit is unavailable, a waiver to conduct the final assessment alongside must be submitted for approval from NTDC(P)" was removed from The Final Assessment Procedure, para 1, to ensure compliance with direction from Command and the intent of the QSP. |
| **28 Jul 20** | It was noted by ADM Mat, that MAR TECHs had been losing the required knowledge to Turn Shafts for a Cold Move and Turn Shafts in Dry Dock. After discussions with the MS Div Cdr, it was decided to add these to criteria to the SEWK Course, to ensure candidates were aware of their duties and responsibilities during these 2 evolutions. The Teaching points, "Cold Moves" and Turning Shafts in Dry Dock" were added to EO 001.02 –Advise Command on Machinery Systems status, Lesson Objective – Describing Operating Restrictions and Parameters, para 1 A, page 3-12. |
| **28 Jan 22** | After consultation with MARTECHs in the fleet, it was determined that there was a conflict of interest having ship's staff conduct the final SEWK evaluation as a system walkthrough. As such, the final evaluation was changed to a board format conducted ashore with NFS as the Convening Authority. The final assessment procedure was adjusted to reflect this change. |

Follow Up Action:

| Item | Action | OPI | Expected Date of Completion |
|---|---|---|---|
| 1 | Refine QSP document | Board Chair/TDO | 06 Aug 18 |
| 2 | Submit recommendation to OA (D Nav P and T) for no trains | TDO | 06 Aug 18 |
| 3 | Develop TEL | MSE Div Comd | 10 Dec 18 |
| 4 | Develop questions and scenarios for the final assessment/board included in the OJPR | MSE Div Comd | 10 Dec 18 |
| 5 | Develop courseware | MSE Div Comd | 10 Dec 18 |

ANNEX D – QSPWB SCALAR



# ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

_____

Name and Initials Rank SN

_____

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. To be employed as a Fuelling I/C aboard a HALIFAX Class frigate, SEWK graduates are required be compliant with SSOs, SEMs and Fueling Orders. To ensure compliance with those Orders, SEWK graduates shall complete the Fueling I/C Checklist at Annex E – Appendix 1 of this QSP.

2. SEWK graduates are not required to complete the Fuelling I/C Checklist as part of the SEWK OJPR to receive the SEWK qualification (ALVA) or challenge the SEWK Board.

3. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

4. As the trainee completes each Performance Objective (PO) to the required standard, the Training PO will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken. The trainee will be provided 180 days to complete the entire OJPR once posted to operational HALIFAX class ship.

5. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO encounters any of these issues they are directed to contact NFS(P) MSE Div or NFS(P) Standards Div for guidance.

PROGESS REVIEW

6. Progress review shall be carried out monthly by the Divisional Officer and/or Section Head.

| OJPR Reviewed By (Rank, Position, Name) | Signature | Date |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

Training PO Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the Sr. Engineering Watchkeeper QSP except as noted and has successfully completed on the job training including all four components of the Qualification Board. The member is qualified to carry out the duties of Sr Engineering Watchkeeper in a Halifax Class vessel and is ALVA qualified.

_____

Engineering Officer Date

ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in this annex shall be used for the following information only:

   a. The unit in which the training and assessment were carried out; and

   b. Any unusual circumstances which preclude compliance with an objective.

| | | | |
|---|---|---|---|
| PO 001 | The primary responsibilities of the Senior Engineering Watchkeeper are to coordinate and control the correct operation of propulsion, ancillary, auxiliary, damage control, power generation and distribution equipment, and systems in all modes of operation, and communicate plant limitations to the OOW in case of an engineering problem or emergency.<br><br>The Trainee shall correctly perform delaying drills to the satisfaction of the MSEO and Sr EWK as per reference C-03-050-HFX/MB-02 (Manual of Emergency Procedures Marine Systems Engineering; at sea).<br><br>Scenarios will include emergency, battle events and operating machinery outside of design parameters which may include the executive order to **Obey Telegraphs.** All drills listed in the manual require a signature. | | |
| EO 001.01 | Advise command on propulsion capabilities and limitations | | |
| | **Description** | **Remark** | **Date/Signature** |
| 001.01.001 | Supervise Main Gearing and Shafting: High Bearing Temperature. | | Sr EWK (PO1/PO2) |
| 001.01.002 | Supervise Main Gearing and Shafting: Interpret bearing flow chart and brief command. | | Sr EWK (PO1/PO2) |
| 001.01.003 | Supervise Propulsion: Shaft Locking - Operate propulsion plant with Shaft Lock | | Sr EWK (PO1/PO2) |
| 001.01.004 | Supervise Propulsion: Pawl Free Operations | | Sr EWK (PO1/PO2) |

| | | | |
|---|---|---|---|
| 001.01.005 | Supervise Propulsion; Normal/Abnormal drive mode operation | | Sr EWK (PO1/PO2) |
| 001.01.006 | Supervise Propulsion Engine Enclosure Fire. | | Sr EWK (PO1/PO2) |
| 001.01.007 | Supervise Propulsion Engine Lube Oil Failure. | | Sr EWK (PO1/PO2) |
| 001.01.008 | Supervise Propulsion Engine High Lube Oil – Bearing temperature. | | Sr EWK (PO1/PO2) |
| 001.01.009 | Supervise Propulsion Engine Fuel Leak. | | Sr EWK (PO1/PO2) |
| 001.01.010 | Supervise Propulsion Engine Over Speed Condition. | | Sr EWK (PO1/PO2) |
| 001.01.011 | Supervise Gas Turbine High PTIT. | | Sr EWK (PO1/PO2) |
| 001.01.012 | Supervise Gas Turbine High Vibration – Compressor Stall above idle. | | Sr EWK (PO1/PO2) |
| 001.01.013 | Supervise Gas Turbine Hot Start. | | Sr EWK (PO1/PO2) |
| 001.01.014 | Supervise Gas Turbine Internal Fire. | | Sr EWK (PO1/PO2) |
| 001.01.015 | Supervise Gas Turbine High Air Intake Differential Pressure. | | Sr EWK (PO1/PO2) |

| | | | |
|---|---|---|---|
| 001.01.016 | Supervise IPMS Failure: Station in Control Crash. | | Sr EWK (PO1/PO2) |
| 001.01.017 | Supervise IPMS Failure: Loss of Critical RTU from Network. | | Sr EWK (PO1/PO2) |
| 001.01.018 | Supervise IPMS Failure: Critical RTU Crash. | | Sr EWK (PO1/PO2) |
| EO 001.02 | Advise command on machinery/system status | | |
| 001.02.001 | Supervise IPMS Failure: Secondary RTU Crash. | | Sr EWK (PO1/PO2) |
| 001.02.002 | Supervise CRPP: CRPP Locking | | Sr EWK (PO1/PO2) |
| 001.02.003 | Supervise Diesel Generator Crankcase: High Pressure Alarm. | | Sr EWK (PO1/PO2) |
| 001.02.004 | Supervise PDE High Fresh Water Temperature. | | Sr EWK (PO1/PO2) |
| 001.02.005 | Supervise Main Lube Oil Failure. | | Sr EWK (PO1/PO2) |
| 001.02.006 | Supervise Diesel Generator Enclosure Fire. | | Sr EWK (PO1/PO2) |

| | | | |
|---|---|---|---|
| As per CFTPOs | Supervise Basin and Main Machinery Power Trials | Basin Trial C-24-293-000/NT-001<br>MMPT C-24-293-000/NT-005 | Sr EWK (PO1/PO2) |
| As per Flash-up Orders | Supervise Flash Up as stated in Flash-up Orders | | Sr EWK (PO1/PO2) |
| As per SSOs | Supervise Machinery/System status on order of Action Stations | | Sr EWK (PO1/PO2) |
| EO 001.03 | Advise command on Power Generation and Distribution system | | |
| 001.03.001 | Supervise Electrical Power Failure: Blackout at Sea. | | Sr EWK (PO1/PO2) |
| Pre-Board requirement | System Description and Operation with Drawing Annex F para 5.a. | | |
| | Gas Turbine (Air Flow) | | MSEC/MPPO |
| | Propulsion Diesel Engine Air Flow | | MSEC/MPPO |
| | Main Gearing | | MSEC/MPPO |
| | Shaftline | | MSEC/MPPO |
| | Controllable Reversible Pitch Propellor | | MSEC/MPPO |
| | Electrical Power Generation and Distribution | | MSEC/MPPO |

| | | | |
|---|---|---|---|
| | Steering | | MSEC/MPPO |
| | Main Lube Oil | | MSEC/MPPO |
| | Fuel Oil Services | | MSEC/MPPO |
| EO 001.04 | Final Practical Assessment (Person in **Bold** is the primary assessor and is final authority for success of candidate) | | |
| 001.04.001 | Final Board | | **Board Chair** |
| 001.04.002 | Abnormal/Battle Scenario | | **Board Chair** |
| 001.04.003 | Safety/Environment Scenario | | **Board Chair** |
| 001.04.004 | Leadership / Control of watch question (MSEO) | | **Board Chair** |

## QUALIFICATION BOARD PROCEDURE

1. The final assessment is to be conducted at the end of the Senior Engineering Watchkeeper training package and will confirm the candidate's understanding of the material studied and the engineering systems they have learned to this point. The purpose of the final assessment is to require the candidate to answer theoretical and practical questions based on battle situations and abnormal operations. Candidate is to display critical thinking ability and explain the correct actions/recommendations in detail in order to advise command. The final assessment is to take place ashore in a board format conducted by NFS(P).

2. To pass the final assessment candidates must successfully challenge a final qualification board. The board will consist of four questions; a major system presentation (see Annex F, Board Format), a drill scenario, a safety and environmental scenario, and a leadership and control of the watch related question.

3. For direction on conduct of the final board, see Annex F.

4. In the event of a final assessment failure or training difficulties, refer to the MAR TECH Sr. Watchkeeper QSP Annex K. A final assessment failure is considered any two failures of an individual component.

# ANNEX E – Appendix 1

## FUELLING I/C CHECKLIST

To be employed as a Fuelling I/C aboard a HALIFAX Class frigate, SEWK graduates are required be compliant with SSOs, SEMs and Fueling Orders.

To ensure compliance with those Orders and perform the job of Halifax Class Fuelling I/C, SEWK graduates shall complete the following Fueling I/C Checklist.

SEWK graduates are not required to complete this Fuelling I/C Checklist as part of the SEWK OJPR to receive the SEWK qualification (ALVA) or challenge the SEWK Board.

Upon completion of this checklist, the MSEC shall fwd a copy of this checklist to the Fomation MSE Chief (or equivalent) for MM Unit Qualification granting.

| **Part B** | **Assessment Act as Fueling I/C and Advise Command** | | |
|---|---|---|---|
| Fueling | As Fueling I/C, Conduct Fueling while at sea | IAW SSOs, SEMs, and Fueling Orders | Snr EWK |
| Fueling | As Fueling I/C, Conduct Fueling while alongside | IAW SSOs, SEMs, and Fueling Orders | Snr EWK |

**<u>CERTIFICATE OF COMPLETION</u>**

_____

Trainee Name and Initials Rank SN

_____

Date Training Completed

I certify that _____ has completed the Halifax Class Fuelling I/C checklist and is employable as a Fuelling I/C aboard a Halifax Class platform.

_____

MSEC Rank, Name, Init. Signature Date

# ANNEX F – FINAL BOARD PROCEDURE

**SEWK Qualification**

1. Prior to granting the Qualification, a final assessment is to be conducted at the end of the Senior Engineering Watchkeeper training package to confirm the candidate's understanding the engineering systems and to ensure they displays the competencies required to perform the role of an SEWK. The final assessment will be conducted in a board format to assess the candidate's answers to theoretical and practical questions based on watch considerations to include abnormal/emergency operations. The candidate is to display critical thinking ability and explain the correct actions/recommendations in detail in order to advise command.

2. The Convening Authority (CA) for the SEWK Qualification board shall be the NFS MS Div Cdrs. The CA will confirm the location, composition, and content of the board, which will normally take place at the Fleet School or other suitable location that can support an uninterrupted and private assessment.

**Board Composition**

3. The board shall be comprised of the following members:

   a. Board Chair: An MS Eng Post-HOD LCdr NTO, with HFX class experience;

   b. Divisional Rep: the candidate's MSEO;

   c. Chief SME: CPO2 MarTech from the mechanical stream post-CHOD tour (ideally the Fleet ChEng, MOC advisor or Training regulator); and

   d. Technical SME: An SEWK or Cert 3 qualified CPO2 or PO1. The technical SME does not have to be from the candidate's unit.

4. While the supporting board members provide insight into the assessment, the Board Chair is ultimately responsible to assess if the candidate has achieved the standard and passed the qualification board.

**Board Format**

5. The Qualification Board shall ensure that the candidate has attained the level of knowledge and possesses the leadership for which an SEWK is expected to demonstrate while on watch. The board shall consist of a:

   a. System description and operation question on one of the following main systems:

      i. Gas Turbine (Air Flow);
      ii. Propulsion Diesel Engine Air Flow;

iii. Main Gearing;
iv. Shaftline;
v. Controllable Reversible Pitch Propellor;
vi. Electrical Power Generation and Distribution;
vii. Steering;
viii. Main Lube Oil; and
ix. Fuel Oil Services.

b. The candidate will be given the main system question and provided with 60 minutes to prepare an answer (drawing) prior to the start of the board.

c. Drill question where the candidate will be expected to demonstrate the proper response IAW the Halifax Class Drill Manual. The underway scenario will be repeated twice for the candidates to familiarize with the drive mode/speed status. The drill Trigger will then be given to the candidate consisting of an Engineering Problem or Engineering Emergency. The candidates are to display critical thinking ability and explain the correct actions/recommendations in detail in order to advise command. The candidates are to react to the changing situation and abnormal operation as the scenario evolves. During the drill response the board will assess the briefs to command and allow command to challenge the candidate's knowledge or recommendations.

d. Safety/environmental question where the candidate will be given time to prepare and present a response on proper actions and advice to Command. The candidate will be given a scenario with a safety/environmental concern while on watch. The candidate will describe the safety/environmental actions to take including immediate actions, briefing the watch, and advising command.

e. Leadership / Control of watch question that will present a leadership challenge to the SEWK. This scenario will be based upon a:

i. Incorrect response to a problem with an auxiliary piece of equipment;
ii. Control of personnel on watch, or a supporting member of the MSED; or
iii. On-watch issue requiring investigation and notification of the MSEO/Ch Eng.

6. All questions and scenarios will be prepared from the question banks provided by NFS(P) MSE Div and NFS(P) Standards Div and held by the Convening Authorities. In order to pass the board, the candidate must receive a mark of at least 60% with the first question having a Pass/Fail criteria for the board (i.e. a failure of this question results in a failure of the board).

## ASSESSMENT FORMAT

7. There shall be a single scoring assessment (whether a checklist, ‘scorecard’ or other method) completed for the board by the Board Chair. The Board Chair shall score the candidate and determine if they PASS or FAIL IAW the rubric provided in Annex O.

a. Other members of the assessment team are to assist the board chair in the assessment by asking relevant questions to the candidate, or otherwise probing the candidate’s abilities. They are to offer their own assessment and recommendations for the board chair’s consideration in scoring and assessment.

b. The practical assessments (oral board and drill) are intended to test the candidate’s ability as Senior Watchkeeper, not simply review what a candidate is already qualified on in previous training (e.g. Engineering Watchkeeper). As such, there shall be a focus on operation of the plant in abnormal modes, adaptation to unusual/difficult circumstances and breakdowns, and the ability to support the intent of command.

c. In the event of a final assessment failure or training difficulties, refer to the MAR TECH Senior Watch keeper QSP Annex K. A final assessment failure is considered any two failures of an individual component.

## AWARDING QUALIFICATION

8. In order to ensure a consistent standard for Quality Assurance purposes the following steps are to be taken for awarding the Senior Watch Keeper qualification:

a. on completion of the board the board chair is to fill out a Board Assessment Form (See Annex O);

b. this form is to be sent and reviewed by the Coastal Training Authorities, FEO - Chief Marine Technician (West Coast)/FEO - Fleet Chief Engineer (East Coast), to ensure that the assessment is consistent with the expected standard. They will recommend to FEO that a candidate be endorsed or not;

c. once satisfied that the expected standard has been achieved, FEO will endorse the qualification and instruct the candidate’s home unit to award the qualification; and

d. NFS(P) MSE or Standards Div will conduct audits of the assessment reports and will also observe some assessments to ensure consistent standards. These audits will happen randomly when the Training Manager raises concerns with a standard, or whenever ship requests assistance in defining/enforcing the qualification standard.

# ANNEX G – ACRONYMS AND ABBREVIATIONS

| Acronym and Abbreviation | Term or Definition |
|---|---|
| ACOS | Automatic Change Over Switch |
| AL | Amendment List |
| CAT | Caterpillar (Diesel Generator) |
| CBRN | Chemical Biological Radiological Nuclear |
| CRPP | Controllable Reversible Pitch Propeller |
| DC | Damage Control |
| DG | Diesel Generator |
| DP | Demo Practice |
| DWAN | Defence Wide Area Network |
| EHM | Equipment Health Monitoring |
| EDI | Explain Demonstration Imitate |
| EWK | Engineering Watch Keeper |
| FT | Field Trip |
| FOS | Fuel Oil System |
| GD | Guided Discussion |
| GL | Guest Lecture |
| GT | Gas Turbine |
| HAZMAT | Hazardous materials |
| HVAC | Heating Ventilation Air Conditioning |
| HPA | High Pressure Air |
| IA | Initial Action |
| IPMS | Integrated Platform Monitoring System |
| IL | Interactive Lecture |
| MCR | Machinery Control Room |
| MCRT | Machinery Control Room Trainer |
| MCOS | Manual Change Over Switch |
| MLO | Main Lube Oil |
| MWM | Motoren-Werke Mannheim (Diesel generator) |
| OBTS | On Board Training System |
| HALOCARBON | Ozone Depleting Substance |
| PDE | Propulsion Diesel Engine |
| PG&D | Power Generation and Distribution |
| LCC | Local Control Console |
| LCP | Local Control Panel |

| | |
|---|---|
| LLMS | Liquid Level Management System |
| LOP | Local Operating Panel |
| LOSCA | Lube Oil Storage and Conditioning Assembly |
| LPA | Low Pressure Air |
| MCC | Machinery Control Console |
| NEM | Naval Engineering Manual |
| NFP | Notice For Power |
| OOW | Officer of the Watch |
| PCOT | Platform Coordination Operations Trainer |
| PT | Power Turbine |
| PTIT | Power Turbine Inlet Temperature |
| SABT | Static Automatic Bus Transfer |
| SSO | Ship Standing Orders |
| SSS (Clutch) | Synchronous Self Shifting |
| Sr EWK | Senior Engineering Watchkeeper |
| SWBD | Switchboard |
| TEL | Technology Enabled Learning |

# ANNEX H - SUGGESTED SEQUENCING AND TIMINGS

| EO | EO Performance Statement | Personnel resources | Required Resources (for scheduling) | Trg Time per resource (mins) | EC Time (mins) | PC Time (mins) |
|---|---|---|---|---|---|---|
| **PO 001: Advise command on marine systems operational capabilities and limitations** | | | | | | |
| EO 001.01 | Advising command on propulsion capabilities and limitations | Instructor | Classroom w/ DWAN MCRT, PCOT, HFX Class ship | 1740 | 360 | 720 |
| EO 001.02 | Advising command on machinery/system status | Instructor | Classroom w/ DWAN MCRT, PCOT | 500 | 360 | 720 |
| EO 001.03 | Advising command on Power Generation and Distribution systems | Instructor | Classroom w/ DWAN MCRT, PCOT | 800 | 360 | 720 |
| EO 001.04 | Advising command on battle damage survey results | Instructor | Classroom w/ DWAN MCRT, PCOT | 1250 | 360 | 720 |
| | | **Actual trg time:** | **6450** | | **Total mins:** | **7422.5** |
| **PT:** | **571.5** | **Course admin time:** | **501** | | **Total days:** | **18.6** |

# ANNEX I – ADDITIONAL TRAINING RESOURCES

1. In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**).

2. The following is a list of the resources identified as being required to conduct the Senior Marine Technician Watchkeeper Course. Some of the resources listed currently exist at the Naval Fleet Schools; however, there are new resources that will be required to maximize the efficiency of training delivery and testing. Purchasing the new items will increase the quality of the student graduating from the training, and significantly reduce the number of required training days ultimately saving time and money over the long-term.

3. The following is a list of classroom requirements:

   a. One large classroom to accommodate maximum number of candidates which include:
      (1) Individual candidate's desks c/w chair;
      (2) Smart Board;
      (3) Dry erase board;
      (4) Projector (power point capable) & screen;
      (5) Blu-ray Player & TV;
      (6) Podium;
      (7) DWAN / LAN connections;
      (8) Laptops/workstations;
      (9) PPS;
      (10) Videos and animations to reinforce engineering knowledge and principles; and
      (11) Student handouts and references.

4. The following is a list of trainers required to support the practical aspects of the training

   a. At NFS's:

      (1) MCRT with EHM data base;

      (2) PCOT with EHM data base; and

      (3) HALIFAX class ship for tours.

   b. Onboard (OJPR):

      (1) OBTS; and

      (2) Ships engineering drill period.

# ANNEX J - COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual FY 2012/13

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

| Course: TDL 300 | | Serial: | | | | |
|---|---|---|---|---|---|---|
| TD | No. Of Students / instructors / | Cost per each | # Days (if applicable) | Initial TE Set-up | Each successive course | 5 Year total |
| Airfare (Return) | 12 | $1,000.0 | | $12,000 | $12,000 | $60,000.00 |
| R&Q | 12 | $36.0 | 20 | $7,920 | $7,920 | $38,600.00 |
| Incidentals | 12 | $17.5 | 20 | $4,200 | $4,200 | $21,000.00 |
| | | | | | | |
| **Subtotal** | | | | **$24,120.00** | **$24,120.00** | **$119,600.00** |
| Material | Quantity | Cost per each | # Days (enter 1 if not applicable) | | | |
| PP&S (Estimate) | 16 | $50.0 | 1 | $800 | $800 | $4,000.00 |
| Classroom (square meters) | 30 | $13.00 | 15 | … | … | … |
| Computer requirements | 16 | $1,000.0 | 1 | $16,000 | $0 | $16,000.00 |
| IT Equip (Proj, Smart board etc) | 1 | $6,000.0 | 1 | $6,000 | $0 | $6,000.00 |
| DWAN connection (Set up) | 1 | $5,000.0 | 1 | $5,000 | $0 | $5,000.00 |
| Instructional Materials (Printed / Handouts etc) | 15 | $30.0 | 1 | $450 | $450 | $2,250.00 |
| **Subtotal** | | | | **$28,250** | **$1,250** | **$33,250.00** |
| PYs (only if course includes incremental on SWE, or course to be delivered off site | Quantity | Cost per day* (SWE if necessary) | # Days (if applicable) | | | |
| N/A | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | **Total Estimated Costs** | | | **$52,360** | **$25,370** | **$152,850** |

# ANNEX K - COURSE HISTORY

1. **This annex shall NOT be deleted**. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

| Reason for amendment (with date) | Details on course history |
|---|---|
| **QSPWB**<br>25 Jun-11 Jul 18 | The Senior Engineering Watchkeeper course was developed to provide training for Sr EWK on a Halifax Class ship. The primary responsibilities of the Sr EWK are to coordinate and control the correct operation of propulsion auxiliary, ancillary, damage control, power generation and distribution systems in all modes of operation, to include sailing through emergency and battle damage events and operation of the plant and engineering machinery outside of design intent. |
| **Prerequisite Amendment**<br><br>23 May 19 | An addition to the course prerequisite, Ch 1, Para 8, includes incorporation of legacy Cert 2E. The EQ applies to a candidate holding the legacy Cert 2E.<br><br>CPO2 Foxe, NPTG(HQ), TM, MAR TECH |
| **OJPR Format**<br><br>9 Aug 19 | OJPR Tasks itemized to reflect training delivery IAW MCS Training.<br><br>CPO2 Foxe, NPTG(HQ), TM, MAR TECH |
| **Steady State Review**<br><br>**02 Mar 23** | QSP reviewed and recommended for steady state approval.<br><br>CPO2 Sutton, NTG(HQ), TM, MAR TECH |
| **OJPR updates**<br><br>**27 Sep 24** | IAW F MSE CPO2 email 10 Jul 24, OJPR updates required, to include:<br><br>1. Installing Shaft locking device and operating propulsion plant with shaft lock;<br>2. Pawl Free GTs;<br>3. CRPP locking;<br>4. Fueling I/C only taught on USQ S2 PO2, they need some real world experience on their package;<br>5. Trial lead for things like a basin trial, MMPT and such;<br>6. Conduct flash up;<br>7. Operate the machinery plant during action stations;<br>8. Operate machinery plant in abnormal drive mode;<br>9. Review of NEM and bearing flow chart;<br>10. Members drawings need to be vetted before a final board. |

| | |
|---|---|
| | The block of Fuelling I/C signature items is best placed as an OJPR for the PO2 RQ course. It really is misplaced in the SEWK QSP. Ideally trg mould be moved from PO2 course to SEWK course, then the SEWK OJPR would align.<br><br>This is an interim solution that should be resolved with the new MARTCH suite of courses, should only need this solution in place for approx. 1 year.<br><br>***LCdr Farmilo, NFS(P) Stds Div Cdr*** |
| **18 October 24** | Secretarial changes made to board rubric after mistakes found by MAR TECH community. |

# ANNEX L ASSESSMENT PLAN

| EC / PC | Scope Covered | Purpose | How | Pass / Fail Criteria | Impact |
|---|---|---|---|---|---|
| EC 001.01 | EO 001.01 | Advise command on propulsion capabilities and limitations | Scenario based assessment including a propulsion problem in MCRT | All critical items,75% of non-critical items | Remedial Training / Re-test |
| EC 001.02 | EO 001.02 | Advise command on machinery/system status | Scenario based assessment including a problem with auxiliary, ancillary equipment problem in MCRT | All critical items,75% of non-critical items | Remedial Training / Re-test |
| EC 001.03 | EO 001.03 | Advise command on power generation and distribution systems | Scenario based assessment including a problem with P,G and D systems in MCRT | All critical items,75% of non-critical items | Remedial Training / Re-test |
| EC 001.04 | EO 001.04 | Advise command on battle damage survey results | Scenario based assessment including battle damage problem in MCRT | All critical items,75% of non-critical items | Remedial Training / Re-test |
| PC 001 | PO 001 | Advise command on marine systems operational capabilities and limitations | Scenario based assessment including multiple equipment problems based on battle damage and/or operating under abnormal conditions in MCRT | All critical items,75% of non-critical items | Remedial Training / Re-test with new scenario / TRB after unsuccessful retest |

| | | | | | |
|---|---|---|---|---|---|
| OJPR | PO 001 | Advise command on marine systems operational capabilities and limitations | Scenario based assessment including multiple equipment problems based on battle damage and/or operating under abnormal conditions in MCRT | 75% | Remedial Training / Re-test with new scenario / TRB after unsuccessful retest |

# ANNEX M  TEL PLAN

| EO | COA 1<br>TEL Delivery | COA 2<br>Interim Delivery | Impact if TEL not implemented<br>- Realized benefits<br>- E/W Coast Delta |
|---|---|---|---|
| EO 001.01 | MCRT/PCOT/OBTS/Video | Ships tour until videos are created | Ships tours not available |
| EO 001.02 | MCRT/PCOT/OBTS/Video | Pictures until videos are created | Use existing TEL |
| EO 001.03 | MCRT/PCOT/OBTS/ Video | Pictures until videos are created | Use existing TEL |
| EO 001.04 | MCRT/PCOT/OBTS/Video | Pictures until videos are created | Use existing TEL |

# ANNEX N TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS

1. The following information was compiled during the QSPWB Sr. Eng WK 25 Jun-11 Jul 18) to give context for developers. It can be removed from the QSP if desired after initial course development.

**<u>Key Target Population Details</u> – Sr. EWK candidates**

2. "The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners" -CFITES Vol 4.

| Characteristic | Target population details |
|---|---|
| **General data** | |
| • Potential group size<br>• Expected annual throughput<br>• Location(s) of the intended learners<br>• Access to computers (job)<br>• Gender<br>• Age<br>• Trades | • Have been posted to a Halifax class ship.<br>• Maximum 12 per course |
| **Learning skills and preferences** | |
| Competencies (general) | • Trade is MARTECH HFX class OFP<br>• 500 hours as EWK, 180 days at sea and 12 months in an operational platform<br>• MS-PLQ qualified |
| Reading and writing level | As per current MARTECH enrolment criteria. |
| Preferences, attitudes towards specific type of learning material and equipment (i.e. TEL, aversion to lectures) | |
| Educational level | As per current MARTECH enrolment criteria. |

| | |
|---|---|
| Previous military training experience | Potential for legacy MSE trades. |
| **Current subject-matter competence** | |
| Experience with target performance | • Will have some exposure to the duties of EWK through observation. |
| Ability to perform any part of the POs | • NA |
| Perception of the impact of achieving the POs on self, work or career | • Uncertain if impact on pay. |
| Job/career stage | • Trainee rank is from MS – PO1.<br>• EWK qualified<br>• MS-PLQ qualified |
| **Social Characteristics** | |
| Relationship to peers | • NA |
| Tendencies toward cooperation or competition | • NA |

# ANNEX O – BOARD QUESTION RUBRIC

| Senior Engineering Watchkeeper – Qualification Board Questions | | |
|---|---|---|
| Candidate: | Ship: | Date: |
| MSEO: | Chief of Department: | PO2/PO1 MAR TECH Mechanical |
| PO2/PO1 MAR TECH Electrical | | |

INSTRUCTIONS

Questions are scored out of 10 pts; 0-8 pts for knowledge / understanding and 0-2 pts for presentation / conduct. A minimum of 60% is required for each of the first two questions in order for the board to carry on past the second question.

KNOWLEDGE / UNDERSTANDING:

0 No knowledge. Incapable of answering main part of the question, including where the system/equipment is located, what it looks like or how to begin even researching the topic.

1 Extremely low knowledge. Incapable of answering main part of the question other than where the system/equipment is located, what it looks like or knows where/who to go to in order to obtain more information.

2 Very low knowledge. Continuous assistance was required to answer the entire main question and answers were insufficient in detail.

3 Low knowledge. Continuous assistance was required to answer half or fewer of the main question and answers were insufficient in detail.

4 Marginal knowledge. Frequent assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

5 Adequate knowledge. Minimal assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

6 Proficient knowledge. Minimal assistance was required to answer the main question and follow up questions in acceptable detail.

7 Strong knowledge. No assistance was required to answer the main question and follow-up questions in acceptable detail.

8 Exemplary knowledge. Answered main question completely and elaborated in detail on specific aspects such that follow-up questions were unnecessary.

PRESENTATION / CONDUCT:

0 Displayed no confidence or professionalism with board members. Answers/diagrams were vague, unorganized and non-committal.

0.5 Displayed poor confidence or professionalism with board members. Answers/diagrams were partially vague and partially unorganized.

1 Displayed confidence and professionalism with board members. Answers/diagrams were organized.

1.5 Displayed strong confidence and professionalism with board members. Answers/diagrams were organized well.

2 Displayed exemplary confidence and professionalism with board members. Answers/diagrams were highly organized.

Senior Engineering Watchkeeper

QUALIFICATION BOARD ASSESSMENT SHEET

| Candidate: | Ship: | Date: |
|---|---|---|

NOTE: Intent is for each assessor to ask one question per section.

| Question | | Content (0-8) | Conduct (0-2) | Score (S) (0-10) | Comments | |
|---|---|---|---|---|---|---|
| System Drawing | 1 | | | | | |
| Drill Scenario | 2 | | | | | |
| Safety / Environmental Scenario | 3 | | | | | |
| Leadership / Control of the Watch Scenario | 4 | | | | | |
| | | | | | Total = | |
| | | | | | RESULT (circle one) = | PASS / FAIL |

Board Member Comments:

Board Member ID (Mandatory):

Name: _____ Signature:_____

| BOARD Average Score | |
|---|---|
| AVERAGE: [ (1)___ + (2)___ + (3)___ + (4)___ ] / 4 = | |



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